

Q4 2024 UPDATE

TowerXchange's Asia guide

A country-by-country guide to
the Asian tower industry

Introduction

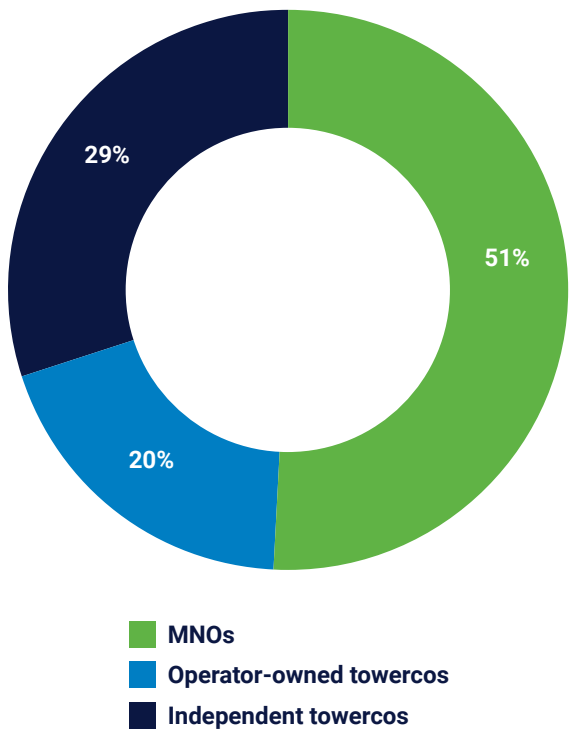
2024 closes with activity continuing across Asia. February’s announcement that PhilTower was merging with MIDC was approved in September, and quickly followed with rumours that Macquarie was looking to monetise its stake in the company, possibly valuing the new entity at US\$700mn to US\$800mn. The creation of the JV has one of the first examples of towerco consolidation in The Philippines and is unlocking new investment the market.

Earlier in January, Bangladeshi towerco Summit Towers snapped up just over 2,000 towers from VEON-owned Banglalink in a in a sale and leaseback deal worth US\$100mn. The telecom market there faced disruption following a coup, but new government ministers and regulators have been appointed and reassured investors.

Japan was a major focus for industry action for the first time. In August 2024, JTOWER accepted an acquisition offer from digital infrastructure investor DigitalBridge. The move has seen its MNO investors cash out and a new and well established international investor link up with the team that brought infrastructure sharing to Japan.

Of course, no summary would be complete without remarking on the turnaround in India. 2023 ended with Ascend Telecom consolidating Tower Vision, and we end the year with news that a consortia of international investors plan to invest in the towerco. The long-awaited

Figure 1: Ownership of Asia’s estimated 1,618,544 towers (excluding China)



Source: TowerXchange

completion of the sale of American Towers sites there to Brookfield ends a period of disruption. Vi is now even planning to roll out 5G in early 2025, a sign of a return to health from one of India’s most troubled operators.

In Malasia, the government has announced a second wholesale network will be created by U Mobile. Malaysia’s smallest mobile operator will be responsible for rolling out a competitive 5G open access network which should drive lease up for existing sites and potentially require U Mobile to bring its small stock of sites to market.

Elsewhere in Asia, smaller deals have also been reported, with Global Tower Corporation in Cambodia raising millions of dollars for rural rollout. But the attention of the industry now turns to two major growth areas.

Firstly, the transformation digital infrastructure networks is creating new opportunities for towercos to add value. To continue to merely provide traditional steel and grass services is no longer enough, energy, in-building, private networks and more are essential new services for towercos to serve their MNO customers.

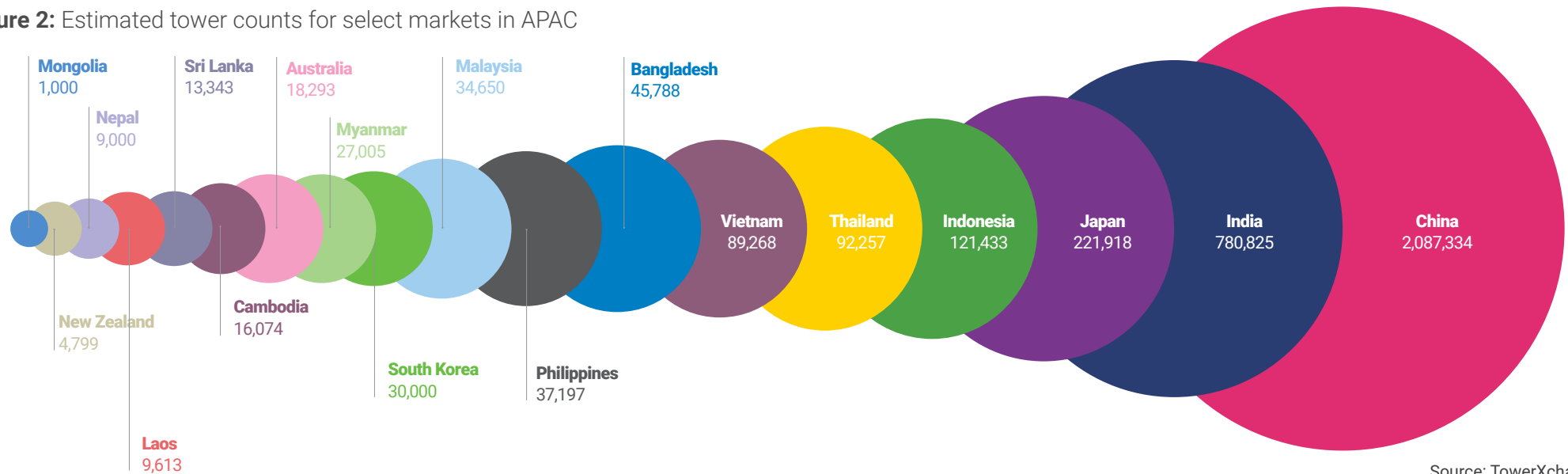
And secondly, 600,000 telecom towers remain in MNO hands in just four markets: Japan, Vietnam, Thailand and India. How these sites come to market over the next few years will offer a major growth opportunity for the industry, if deals can be struck.

David Tran has joined TowerXchange as our new Head of Asia, to provide information for our guide please email david.tran@towerxchange.com



INTRODUCTION

Figure 2: Estimated tower counts for select markets in APAC



Source: TowerXchange

Figure 3: SIMs per tower in select countries across APAC

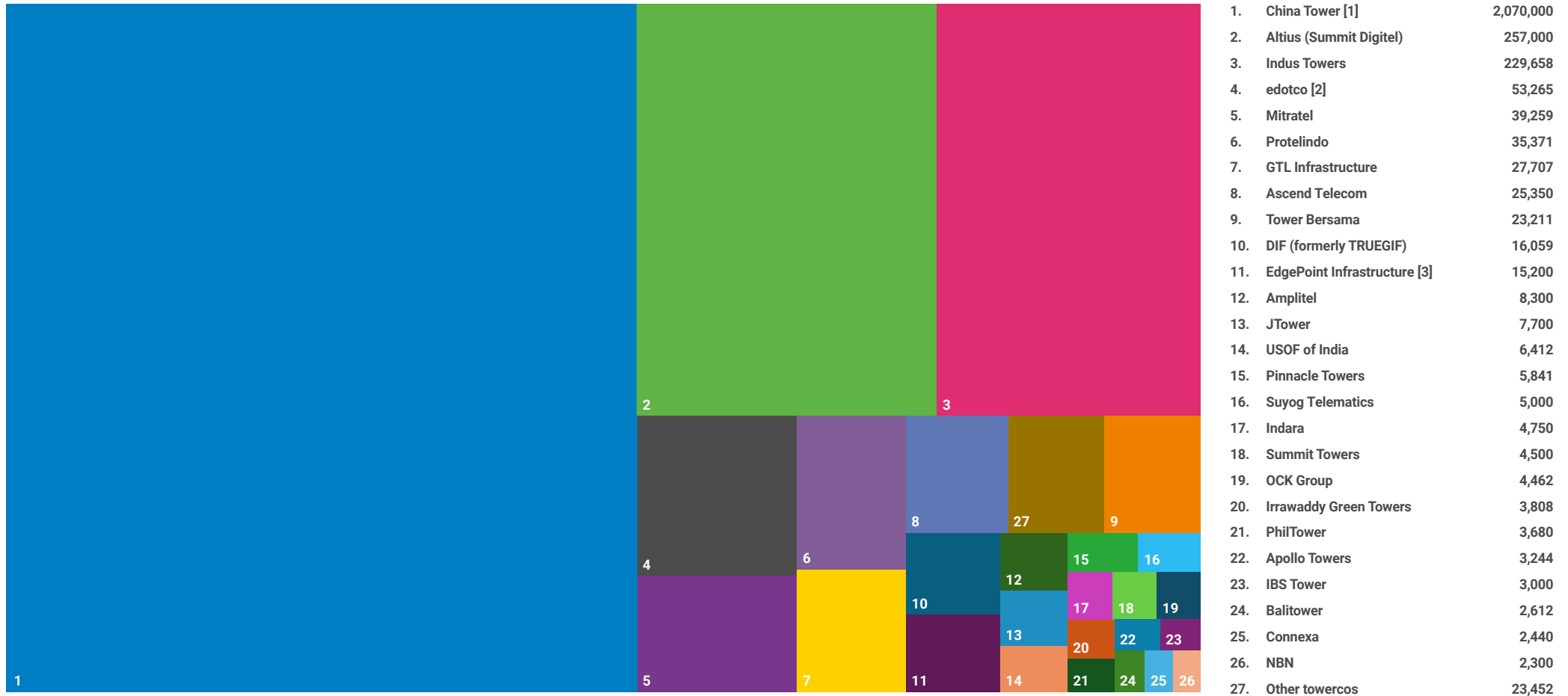


Source: TowerXchange



INTRODUCTION

Figure 4: Tower counts of Asia's towercos



[1] China Tower Corporation area reduced to better illustrate the relative market share of the other towercos. If illustrated to scale, CTC's areas would be 3.2x the rest of the industry combined.

[3] EDOTCO count includes both owned and managed sites.

[4] EdgePoint Infrastructure's site count includes Centratama (10,750).

**Figure 5:** Major tower transactions in the Asian tower market

Year	Country	Seller	Buyer	Towers	Value (US\$)	Value per towers (US\$)	Deal Structure
2024	Japan	JTOWER	Digital Bridge	7,700	630,000,000	81,818	Company acquisition
2024	Philippines	Globe Telecom	Frontier Tower Associates	1,037	235,216,000	226,823	Sale & Leaseback
2024	Australia	Vertel	Everest Infrastructure Partners	200			Sale & Leaseback
2024	Bangladesh	Banglalink	Summit Towers Limited	2,012	100,000,000	49,701	Sale & Leaseback
2024	Philippines	Globe Telecom	MIESCOR Infrastructure Development Corporation (MIDC)	2,180	465,050,000	213,325	Sale & Leaseback
2024	India	American Tower	Brookfield	78,300	2,500,000,000	31,928	Company acquisition
2023	Indonesia	PT Gametraco Tunggal	Miratel	803	112,000,000	139,476	Company acquisition
2023	India	Tower Vision India	Ascend Telecom Infrastructure	10,000			Company acquisition
2023	Philippines	Globe	Unity Digital Infrastructure	447	97,000,000	217002	Sale & Leaseback
2023	Philippines	Smart	Frontier Tower Associates	1012	220,000,000	217391	Sale & Leaseback
2023	Indonesia	IOH	Mitratel	997			Sale & Leaseback
2022	New Zealand	2Degress	Connexa	1124	1,080,000,000	960854	Sale & Leaseback
2022	Philippines	Smart	Unity Digital Infrastructure	650	165,000,000	253,846	Sale & Leaseback
2022	Philippines	Globe	Philtower Consortium	1350	338,000,000	250,370	Sale & Leaseback
2022	Indonesia	Telkomsel	Mitratel	6000			Sale & Leaseback
2022	Philippines	Globe	Frontier Tower Associates	3,539	810,000,000	228,878	Sale & Leaseback
2022	Philippines	Globe	MIDC	2180	468,000,000	214,679	Sale & Leaseback

**Figure 5:** Major tower transactions in the Asian tower market

Year	Country	Seller	Buyer	Towers	Value (US\$)	Value per towers (US\$)	Deal Structure
2022	Philippines	Alliance Towers Corporation	Bersama Digital				Company acquisition
2022	New Zealand	Vodafone New Zealand	InfraRed Capital Partners and Northleaf Capital Partners	1484	1,100,000,000	741240	Sale & Leaseback
2022	New Zealand	Spark	OTPP	1264	900,000,000	712025	Sale & Leaseback
2022	Australia	Stilmark	OMERS Infrastructure	70			Company acquisition
2022	Australia	TPG Telecom	OMERS Infrastructure	1237	674,000,000	544,867	Sale & Leaseback
2022	Philippines	PLDT	EDOTCO Group	2973	798,700,000	268,700	Sale & Leaseback
2022	Philippines	PLDT	EdgePoint Infrastructure	2,934	671,300,000	228,800	Sale & Leaseback
2022	Australia	Axicom	Australia Tower Network (ATN)	2001	2,680,000,000	1,339,330	Company acquisition
2022	Japan	NTT	JTower	6002	870,000,000	144952	Sale & Leaseback
2022	Indonesia	Axiata	EDOTCO	1021	52,200,000	51,126	Sale & Leaseback
2021	Malaysia	Touch Mindscape	EDOTCO	1000	400,000,000	400,000	Company acquisition
2021	Indonesia	STP Tower	Protelindo	6700	1,170,000,000	174,626	Company acquisition
2021	Indonesia	Telkomsel	Mitratel	4,000	400,000,000	100,000	Sale & Leaseback
2021	Japan	NTT	JTower	71			Sale & Leaseback
2021	Indonesia	Indosat Ooredoo	EdgePoint Infrastructure	4,247	750,000,000	178,571	Sale & Leaseback
2021	Malaysia	Asiaspace	Edgepoint Infrastructure	178			Company acquisition
2021	Indonesia	Centratama	Edgepoint Infrastructure	3800			Company acquisition

**Figure 5:** Major tower transactions in the Asian tower market

Year	Country	Seller	Buyer	Towers	Value (US\$)	Value per towers (US\$)	Deal Structure
2020	Indonesia	XL Axiata	Protelindo	1,728	164,000,000	94,907	Sale & Leaseback
2020	Indonesia	XL Axiata	Centratama	1,054	118,260,000	112,201	Sale & Leaseback
2020	Indonesia	Telkomsel	Mitratel	6,050	700,000,000	115,702	Sale & Leaseback
2020	Indonesia	IBS Tower	Tower Bersama	3,000	280,000,000	93,333	Asset purchase
2019	Indonesia	Indosat Ooredoo	Mitratel	2,100	312,750,000	148,929	Sale & Leaseback
2019	Indonesia	Indosat Ooredoo	Protelindo	1,000	137,250,000	137,250	Sale & Leaseback
2019	Thailand	True Move	Digital Telecommunications Infrastructure Fund	778	512,000,000	-	Sale & Leaseback
2019	Laos	MTCL	EDOTCO		1,460,000		80% stake
2019	Indonesia	PT Persada Sokka Tama (PST)	Mitratel	1,589			Company acquisition
2018	Cambodia	Seatel	EDOTCO	345			Sale & Leaseback
2018	India	Idea Cellular	American Tower	9,900	592,700,000	59,868	Sale & Leaseback
2018	India	Vodafone	American Tower	10,200	592,900	58,127	Sale & Leaseback
2018	India	Reliance Communications	Reliance Jio	43,000			Sale & Leaseback
2018	Thailand	True Move	Digital Telecommunications Infrastructure Fund	2,589			Sale & Leaseback
2018	Indonesia	GOLD	Tower Bersama	305			51% stake
2018	Malaysia	Tanjung Digital Sdn Bhd	EDOTCO	225			80% stake
2018	Myanmar	PAMEL	TPG	1,300			Company acquisition

**Figure 5:** Major tower transactions in the Asian tower market

Year	Country	Seller	Buyer	Towers	Value (US\$)	Value per towers (US\$)	Deal Structure
2018	Indonesia	KIN	SMN	1,400			Company acquisition
2018	Indonesia	Providence (KIN)	Protelindo	1,400			Company acquisition
2017	Indonesia	STI	IBS Tower	371	31,000,000	83,557	Sale & Leaseback
2017	Australia	Southern Cross Austereo	Axicom	56	9,500,000	169,643	Sale & Leaseback
2016	Indonesia	XL Axiata	Protelindo	2,500	250,000,000	100,000	Sale & Leaseback
2016	Vietnam	VNI (SEATH)	OCK Group	1,972	50,000,000	25,355	Company acquisition
2016	India	Viom Networks	American Tower	42,200	1,180,000,000		Sale & Leaseback
2015	Myanmar	Digicel MTC	EDOTCO	1,250	221,000,000		Sale & Leaseback
2015	Australia	Crown Castle	MIRA-led consortium	1,772	1,600,000,000	902,934	Company acquisition
2015	India	KEC International	American Tower	381	13,000,000	34,121	Company acquisition
2014	Indonesia	XL Axiata	STP	3,500	460,000,000	131,429	Sale & Leaseback
2014	Malaysia	KJS	YTL Power International	309	15,000,000	48,544	Company acquisition
2013	Indonesia	Hutchison	STP	300	68,000,000	226,667	Sale & Leaseback
2012	Indonesia	Hutchison	Protelindo	503			Sale & Leaseback
2012	Indonesia	Indosat	Tower Bersama	2,500	519,000,000	207,600	Sale & Leaseback
2012	Indonesia	PT Central Investido	Protelindo	152			Company acquisition
2011	Indonesia	Infratel	Tower Bersama	595			Company acquisition



Australia

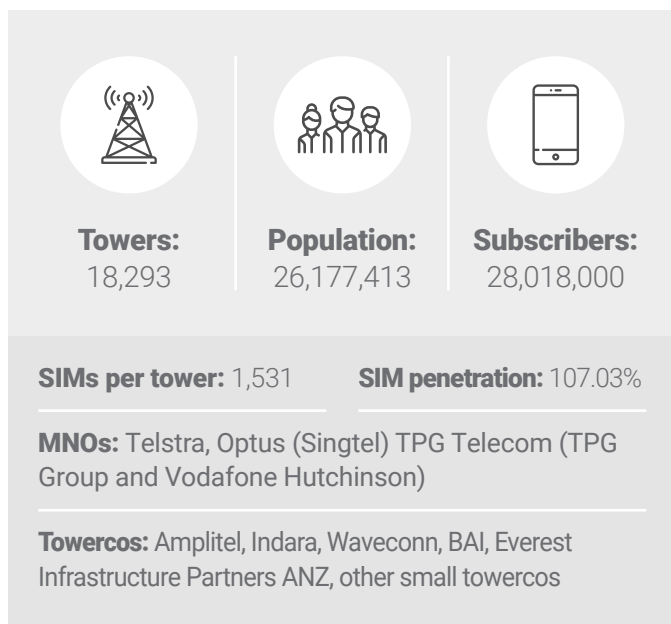
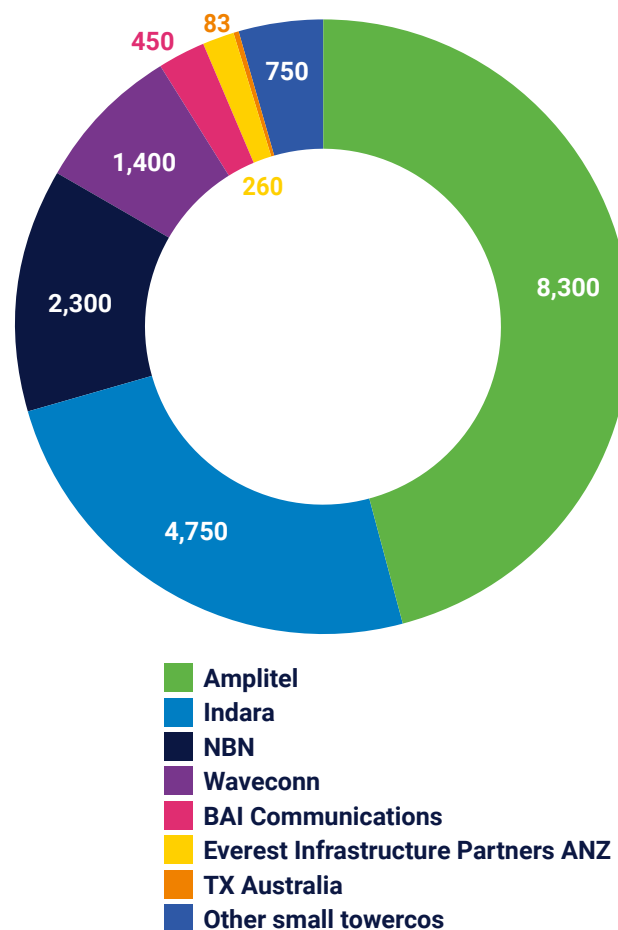


Figure 6: Australia – estimated tower count



Source: TowerXchange

The towers currently host Vertel's network equipment rather than a GSM operator's equipment like a traditional towerco, but the opportunity to lease up the sites and their strategic location makes them an attractive acquisition. Everest Infrastructure Partners are an interesting acquirer for the telecom towers as their traditional business focus on land under towers.

Waveconn is also to acquire American Tower's Australian property business, adding a further 170 sites to its portfolio comprising land under mobile towers and building rooftops. The transaction is expected to close in Q4 2024.

Australian Prime Minister Anthony Albanese's-led government also seems to be recognising the importance of establishing more reliable connectivity across Australian society, which is set to facilitate build activity and increased resiliency in existing infrastructure. The Federal Government's amended Telecommunications in New Developments policy, published in February 2024, is set to reshape the landscape for property developers. The updated policy now includes provisions for mobile telecommunications infrastructure, specifically targeting residential developments with more than 50 lots.

Australia's towerco landscape presents itself a solid picture of tower growth and consolidation, backed by measures from the Federal Government to support, in particular, rural connectivity and best maintain critical passive infrastructure. Everest Infrastructure ANZ completed the acquisition of towers sites owned by Australian telecommunications carrier and ICT managed services provider Vertel in May 2024. TowerXchange estimates Vertel had 200 towers, which will increase Everest's tower count to around 260 in Australia.



AUSTRALIA

Buoyant developments in the early half of 2024 year followed a year of consolidation in Australia in 2023, and after 2021 and 2022 which saw all three MNOs selling the remainder of their towers.

Amplitel was formed by Telstra when they sold a 49% stake in their 8,200 tower portfolio to an investment consortium consisting of Australia's sovereign wealth and pension funds, including Future Fund and Commonwealth Superannuation Corp.

Indara was the second new towerco to be formed in 2021. It started its life in October as Australian Tower Network (ATN), when Singtel owned Optus sold a 70% stake in its 2,312 tower and rooftop sites to AustralianSuper.

The company rebranded to Indara in October 2022. In addition to leasing these sites from Indara, Optus will serve as anchor tenant on 565 new build-to-suit towers that was set to be constructed before the end of 2024. However, AustralianSuper is widely reported to have fallen out with Optus over the new build of towers.

In May 2022 TPG Telecom announced they had sold a 100% stake in their 1,237 ground-based and rooftop sites to Canadian pension fund OMERS for \$US670 million. Unlike the other tower sales there was no formal auction process for the TPG towers.

In June 2022, OMERS Infrastructure announced

its second tower deal in as many months with an acquisition of Stilmark Group. During the Telstra and Optus transactions, OMERS and Stilmark established joint venture Symphony Consortium, with the intention of combining Stilmark's tower expertise with the capital available to OMERS. OMERS closed both deals in August 2022 and rebranded their two new portfolios to form a towerco called Waveconn.

Indara continued their M&A activity with the acquisition of smart street furniture designers and small cell HUB and ENE HUB in September 2022. Smaller players in the Australian market include BAI Communications, who own a portfolio of around 600 towers, but have focused their attention more on urban connectivity and subway networks on a global scale as of late. However, one of the influential figures driving in Australia's towerco revolution, Cameron Evans, announced his resignation as CEO of Indara in November 2024, leading to a future change at the helm of the company.

In February 2022 it was announced that BAI's Australia tower business was up for sale for \$US1.4 billion, but a transaction has yet to be announced.

The remainder of the towers in the country are held by a handful of smaller independent towercos. Just under 2,500 towers have been erected by NBN - the Government owned national broadband network, although currently all of these towers are home to fixed wireless rather than mobile technology. The world's

largest independent towerco, American Tower, has also quietly established a presence in the Australian market through their acquisition of InSite Wireless, a lease aggregator company.

In the era of MNO owned towers, the Australian federal and state governments displayed their commitment to maintaining and improving telecoms infrastructure to keep the country connected. Australia's Minister of Communications said that more than 100 communities and locations across rural and regional Australia will benefit from significant communications upgrades from the programs in an effort to improve mobile coverage issues, deliver fibre upgrades and fund public Wi-Fi.

Despite a financial commitment to improving mobile service in rural Australia, the ACCC blocked a revolutionary "Multi-Operator Core Network" commercial agreement proposed between Telstra and TPG Telecom. Under the terms of the agreement, Telstra were planning to use TPG held spectrum to boost capacity in rural locations, while TPG were to decommission around 700 rural towers, instead relying on access to around 3,700 Telstra sites to serve its subscribers, boosting its coverage. The proposal was rejected following anti-competition claims from Optus.

Bangladesh



Towers:
45,788



Population:
171,186,372



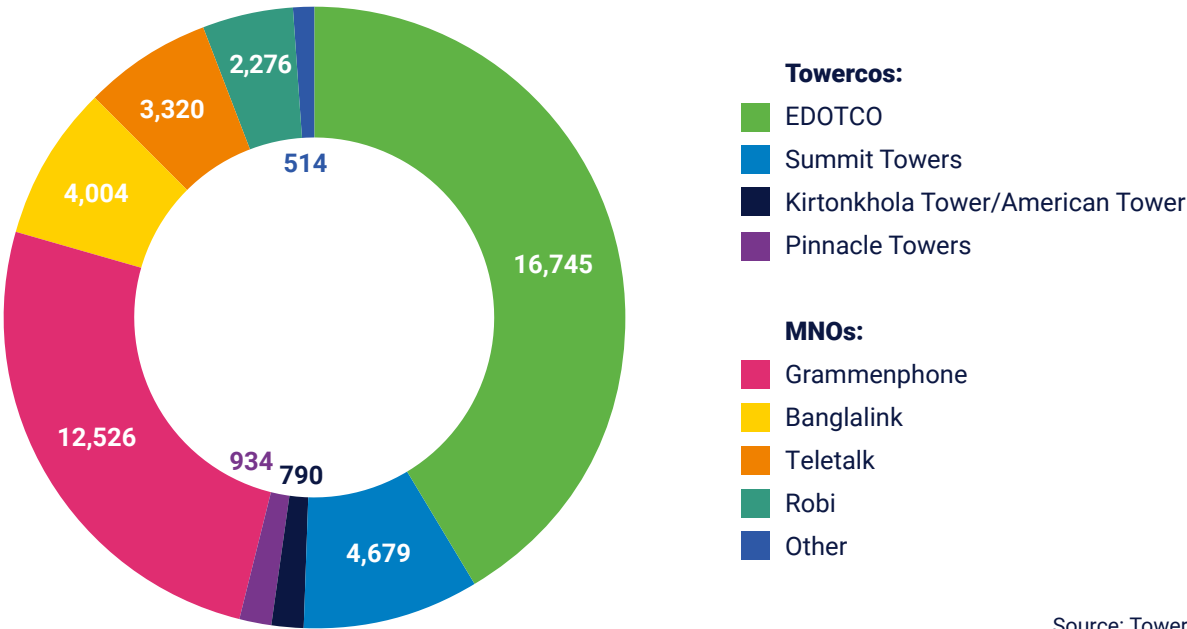
Subscribers:
180,198,049

SIMs per tower: 3,689
 SIM penetration: 105.26%

MNOs: Grameenphone (Telenor), Robi (Axiata), Teletalk (Government of Bangladesh), Banglalink (VEON)

Towercos: EDOTCO, Summit Towers, Kirtonkhola Tower Bangladesh, Pinnacle Towers

Figure 7: Bangladesh – estimated tower ownership



Source: TowerXchange

Despite a country rocked by nationwide protests in a coup that brought to the end of the tenure of its Prime Minister in August 2024, Bangladesh’s tower industry is bouncing back. With an interim government in place, the focus since has been to establish stability in the country and make infrastructure a priority. This has given towercos the confidence to progress with

projects and network expansion. The recent tax breaks on renewable energy sources like solar power, have also made it cheaper for us to switch to cleaner energy, highlighting the government’s proactive approach.

Bangladesh’s towerco market experienced seismic activity rolling into 2024 when VEON-owned MNO

Banglalink relinquished part of its tower portfolio, offloading 2,012 towers in a sale and leaseback transaction to Summit Towers for \$US100m. The deal will mean Banglalink will be provisioning tower infrastructure services for an initial 12 years, in the process crystallising VEON and Banglalink’s position as an asset-light digital operator, and asserts Summit



BANGLADESH

Tower's position as a major digital infrastructure provider in the country. The transaction alters the towerco landscape in a country where it operates a four MNO market consisting of state-owned Teletalk, market leader Grameenphone, who account for 46% of mobile subscribers, Axiata subsidiary Robi and Banglalink.

One of the MNOs, Banglalink, most recently launched a network sharing programme with Teletalk. It will see Teletalk subscribers seamlessly switch to Banglalink's network, utilising Banglalink's approximate network of 12,000 towers. The move could create momentum in the country for active sharing of tower infrastructure.

Towerco licenses were first issued back in November 2018 to incumbent player EDOTCO, Summit Communications' Summit Towers, American Tower-owned Kirtonkhola Tower Bangladesh and AB HighTech Group, which was acquired by KKR-backed Pinnacle Towers in October 2022.

In March 2020, after much negotiation, the Bangladesh Telecommunication Regulatory Commission (BTRC) finalised the long-awaited Service-Level Agreement (SLA) that enabled licensed towercos to start operating in the country by providing firm guidelines on how MNOs and their infrastructure counterparties should engage commercially.

During the negotiation period of the regulation new builds all but stopped across the country which has led

to significant built up demand that has enabled high levels of organic growth for the new tower companies.

In terms of the scope of services towercos can provide, some of the legislation is open to interpretation. License owners can own and maintain passive infrastructure with this covering power (primary power is provided as a passthrough with towercos providing battery back-up). In-building solutions are not explicitly mentioned, but towercos are able to share existing infrastructure and so could rent out systems. Bangladesh provides an excellent market for a towerco to achieve fast and significant growth.

Current estimates predict that around 2,500 Grameenphone sites are shared by at least a second operator. After the November 2023 deadline, the MNO that owns the tower must sublet to a towerco that in turn rents space to the co-locating operator.

In terms of operating conditions, the rainy season demands exceptional cell site autonomy which makes Bangladesh a key market for energy vendors, particularly those that can help with energy storage.

Bangladesh has the third worst access to electricity in rural areas of the Asian countries that TowerXchange tracks, so securing access to power on these bad and off-grid sites will be a key priority as coverage increases. Estimates put the number of completely off-grid sites at around 10%, while bad grid that is so unstable it might as

well be treated as off-grid for the purposes of ensuring uptime boosts this number closer to 25%.

EDOTCO own or manage nearly 19,000 sites, the majority of which were inherited from sister company Robi following the transfer of assets.

The country's MNOs have also been holding conversations with Bangladesh's three other licensed towercos in a bid to diversify from relying solely on EDOTCO. Banglalink's divestment in tower assets is very much a case in point of the seemingly cooperative relationship between MNOs and tower operators.

In October 2022, AB HighTech was acquired by KKR-backed Pinnacle Towers, who run Frontier Tower Associates in the Philippines. MNO have long set their stool out to reduce their tower counts.

Kirtonkhola Tower was awarded a 300 tower BTS contract by Banglalink in 2021 and currently owns over 500 towers. Meanwhile, Telenor and Grameenphone have launched a process to sell 12,000 of Grameenphone's towers.

EDOTCO has been declared by the BTRC to have significant market power and has been limited to acquiring a maximum of 25% of any towers sold by MNOs. As such, competition will be fierce between Summit, Pinnacle and American Tower to become the first towerco to approach EDOTCO's scale.



Cambodia



Towers:
16,084



Population:
16,767,842



Subscribers:
19,505,294

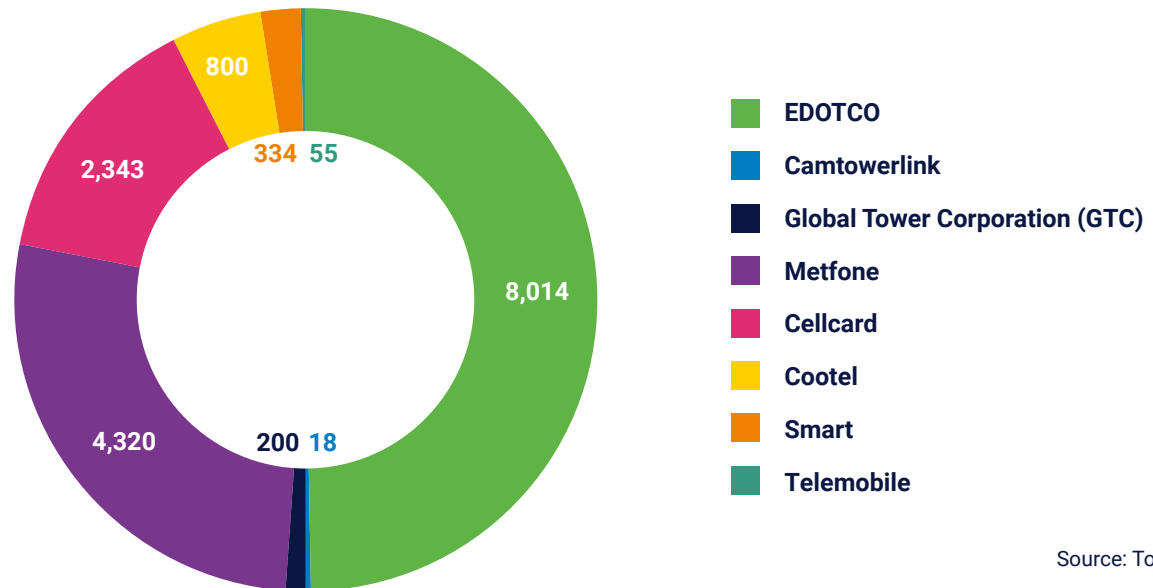
SIMs per tower: 1,213

SIM penetration: 116.33%

MNOs: Smart (Axiata), Cellcard (CamGSM), Metfone (Viettel) Seatel and Cootel

Towercos: EDOTCO, Camtowerlink, Telemobile, Global Tower Corporation

Figure 8: Cambodia – estimated tower ownership



Source: TowerXchange

There are five MNOs in the Cambodian market, Cellcard (owned by Cambodian CamGSM), Smart (owned by Axiata Group), Metfone (owned by Viettel) and Chinese backed Seatel and Cootel. Cellcard, Smart and Metfone account for 90% of Cambodia's 21.4mn subscribers (in a population of 16.7mn). In June 2023, Cellcard launched the process to hold the largest IPO in Cambodian history, and aims to use the funds it

will raise to further expand its network and services, as well as to repay some of its debt. The MNO has carved out 2,300 towers into a new entity by the name of Telemobile. Initially, Telemobile will take over management of Cellcard's towers and will also provide energy. The portfolio is a combination of rooftop and macro sites. Telemobile will be working with an ESCO to manage power in the future.

4G was first launched in Cambodia in 2016 and whilst the country's leading MNOs have been carrying out 5G trials following an agreement with Huawei to rollout 5G infrastructure, spectrum in the country is yet to be awarded. Regulators were concerned that there would be inefficiencies and waste if all five MNOs built their own 5G infrastructure, so withdrew the licenses that the operators had been using for their 5G trials.



CAMBODIA

While this is frustrating, the government's willingness to step in to encourage infrastructure sharing could pose an opportunity for towercos who can support co-location services and the sought after efficiencies.

Historically, around 200-300 new sites have been built in the country per year. There are now four towercos active in the Cambodian market. EDOTCO entered in 2013 and further bolstered its position with the acquisition of 325 sites from Seatel in 2018. News broke in August 2021 that EDOTCO had entered into a new agreement to provide 500 sites for Metfone. The majority of their sites prior to this were built or acquired for Smart Axiata.

EDOTCO manage 800 sites for Cooltel and 334 for Smart. MNOs in Cambodia including Axiata maintain responsibility for power at their sites. EDOTCO is joined by local player Camtowerlink – whose sites are all located around the famous Angkor Wat UNESCO world heritage site, following a partnership with the landmark temple's custodians.

Global Tower Corporation (GTC), another towerco, owns and manages 114 sites. Competition for sites in Cambodia will continue to be fierce as tower builders strive for site and 4G network densification of sites in the largely populated cities for the operators.

In October 2024, Global Tower Corporation signed a financial agreement with the Export-Import Bank of Malaysia Berhad securing US\$19.97mn to build and operate 400 towers across Cambodia.

As dictated by the regulation, the towercos will all have to avoid erecting sites in the same location as other towers, to reduce the parallel infrastructure deployed in the country.



India



Towers:
777,585



Population:
1,417,173,173



Subscribers:
1,142,930,000

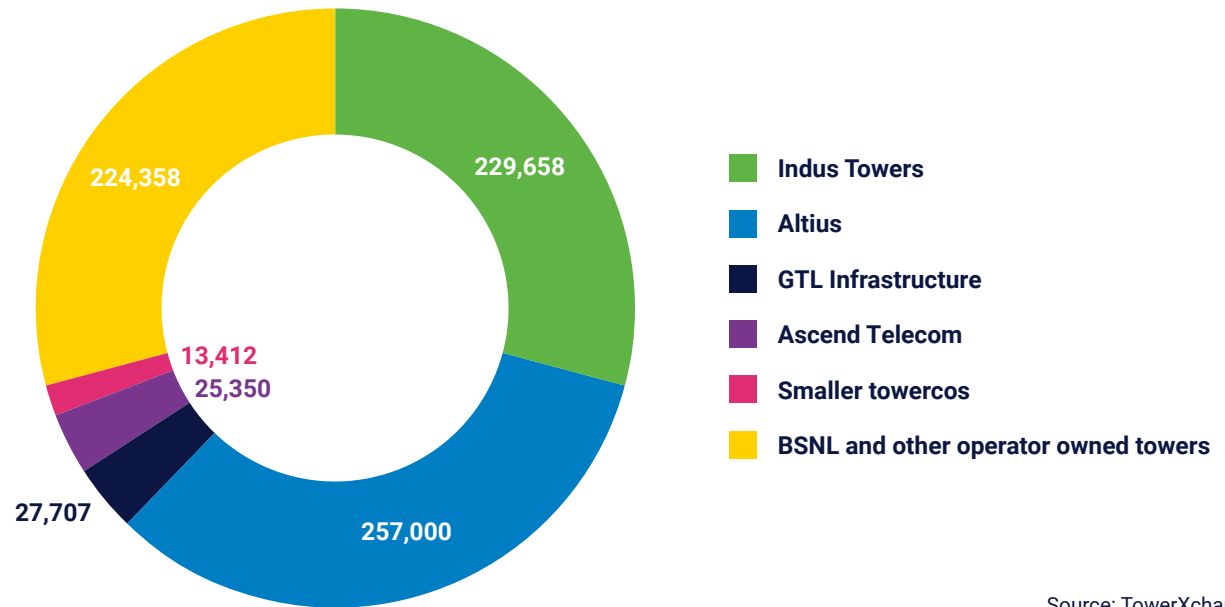
SIMs per tower: 1,463

SIM penetration: 80.65%

MNOs: Reliance Jio, Bharti Airtel, Vodafone Idea, BSNL

Towercos: Indus Towers, Altius, American Tower, GTL Infrastructure, Ascend Telecom, USOF, Suyog Telematics, Railtel, Signotox, Roam Digital

Figure 9: India – estimated tower ownership



Source: TowerXchange

2024 represents a defined year towards rationalisation of India's telecom tower market. American Tower's exit and improved 5G deployments will help ease the pain of the last year.

American Tower officially exited its operations in the country in September 2024, having agreed a deal with Summit Digital's owners Brookfield (operating under

the new brand name Altius) to sell their loss-making 77,712 sites for US\$2.5bn at the beginning of the year. By acquiring American Tower Brookfield may be creating further opportunities for lease-up for its two businesses (Brookfield's DIT (Brookfield's infrastructure investment trust) itself houses Brookfield's other telecom tower and related infrastructure investments in India: Summit Digital, Crest Digital and the newly launched Roam Digital).

Brookfield's anchor client is Reliance Industries' Jio, whose entrance into the market caused 14 operators to cease operations between 2016 and 2021.

This led to the abandonment of thousands of towers and significant decommissioning which impacted the profitability of the majority of the country's towercos. The market leader is Indus Towers, which reached its



INDIA

180,000 plus site count following a merger with Bharti Infratel which was completed in November 2020. Bharti Infratel was the carve out towerco of Bharti Airtel, and they also owned a 42% stake in Indus Towers pre-merger, alongside Vodafone. Summit Digitel is owned by Brookfield, who purchased Reliance Jio Infratel's tower assets a few months prior in September 2020.

Vodafone Idea will be introducing 5G services in India from March 2025. Its launch is three months later than initially planned, but indicates a return to financial health after several years of retrenchment. Plans have been pushed back in order give the operator more time to execute spectrum rollout in 17 different metros and cities, and to improve its 4G coverage.

In June 2024, Vodafone Group sold 18% of its shares in Indus Towers for €1.7bn (US\$2.16bn) so it can repay its debts in relation to the outstanding bank borrowings of €1.8bn secured against Vodafone's Indian assets. Therefore the transaction is unlikely to benefit Vi's own debt situation.

Other independent towercos include GTL Infrastructure, who were rocked especially hard by the abandonment of towers and Ascend Telecom, which has reportedly approached by KKR, Stonepeak Partners and Macquarie Capital. A few operators retain towers in the country, BSNL, RCOM, Airtel and Bharti Hexacom all retain some towers.

Following Jio's entrance and the rapid consolidation, the Indian market now functions with four MNOs. Jio and Bharti

Airtel lead the pack, while Vodafone Idea lags behind in terms of network coverage. BSNL has a small market share.

The more stable status quo has allowed for 5G spectrum auctions to be successfully completed. Airtel and Jio have taken adequate spectrum to meet the requirement for their 4G and 5G networks across all 22 telecom circles. Vi look likely to continue to focus on 4G development and be more strategic with its 5G rollout. Initially it looks like they will focus on main urban locations.

All three MNOs rolled out their network in a phased manner in 2022, starting with the major metro cities, state capitals and other Class A cities. Jio and Airtel have made significant progress and are expected to reach nationwide coverage in the next two years.

BSNL is focusing on completing its 4G deployment and initially did not participate in the auction, although in early 2023, it requested additional spectrum beyond what was reserved for it in 2022.

The demand for new sites driven by 5G is expected to revitalise the tower industry, with estimates that India needs to add 500,000 new sites in the next few years, although many of these will be smaller infill solutions rather than macro sites.

New forms of wireless infrastructure including small cells, poles and DAS are being deployed across the country

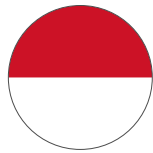
at scale. In August 2022, BSNL announced plans to sell 10,000 towers worth approximately US\$500mn. The sites for sale are those where BSNL shares infrastructure with one of India's three private MNOs, implying a strong tenancy ratio across the portfolio.

Power remains a key topic for the Indian towerco market, where towercos provide power as a service. India recently transformed to an energy surplus country and the government has put a lot of effort into encouraging renewables such as solar, wind, hydro and hydrogen power solutions as well as improving the quality and availability of the grid.

The government's mission to create 100 Smart Cities and Towns in India is also a driver of growth for the country's infrastructure players. The initiative was launched in 2015 and to date, around two thirds of the identified 7742 projects have been completed.

To support this rollout Telecom Minister Ashwini Vaishnaw explained that BSNL was in the process of ordering 6,000 towers and would shortly follow up with two additional orders; one for a further 6,000 and one for 100,000.

BSNL have carved out their 159,000 towers into a separate entity, although TowerXchange understands that no external capital has been invested, and the company does not operate as a towerco in terms of actively marketing and seeking co-locations at sites. As such the towers are still classified as being owned by the MNO.



Indonesia

Towers:
121,463

Population:
275,501,339

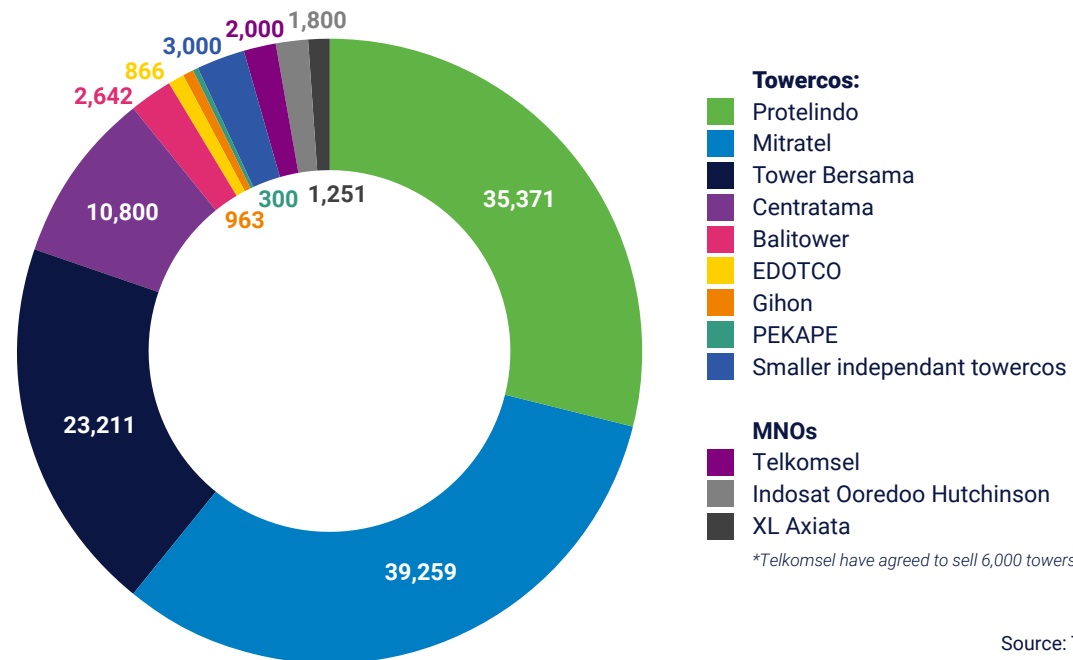
Subscribers:
316,552,572

SIMs per tower: 2,606 **SIM penetration:** 114.90%

MNOs: Telkomsel, XL Axiata, Indosat Ooredoo Hutchison, Smartfren

Towercos: Mitratel, Protelindo, Tower Bersama, Centratama (EdgePoint Infrastructure), IBS Tower, Balitower, EDOTCO, Gihon, Persada Sokka Tama, PEKAPE.

Figure 10: Indonesia - estimated tower ownership



Source: TowerXchange

Indonesia is a fascinating market to study from a towerco perspective, as one of the oldest, most established and yet most dynamic markets. Over the past decade, the country's mobile network operators have been monetising their towers through tower carve outs and sales and have relied heavily on towercos for the rollout of new sites. A transaction in early 2023 between IOH and Mitratel marked

the end of an era, as once the deal closes towercos will own 100% of the nation's sites.

In addition to building new sites and acquiring MNO owned towers, the country's leading towercos have further bolstered growth through the acquisition of smaller towercos making for one of the most dynamic tower M&A markets in the world.

Protelindo's fibre subsidiary iForte acquired IBS Towers, increasing its fibre network to over 170,000km in order to strengthen its fibre to the tower (FTTT) and fibre to the home (FFTH) businesses, along with its 3,000 towers.

Prior to the entrance of DigitalBridge backed EdgePoint Infrastructure in late 2020, there was an expectation that this



INDONESIA

consolidation would continue until the market was composed of Mitratel, Protelindo and Tower Bersama.

However, a further step was made in the opposite direction in March 2022 as pan-Asian powerhouse EDOTCO made its anticipated move into Indonesia with an acquisition of 1,021 towers from parent company Axiata. Of these towers 859 are owned and 162 are managed by EDOTCO. The towers were priced at US\$52.2 million.

In June 2022 it was reported that Protelindo were looking to raise \$US1 billion by selling between 15% and 20% of the business. Protelindo's 2015 acquisition of iForte provided the springboard for the company to grow its fibre footprint and use it to support 3 different business lines, FTTT, FTTH and Connectivity. Protelindo also bought out PT Solusi Tunas Pratama (STP Tower) in 2021, which saw them inherit 6,700 towers and over 9,300km of fibre.

Mitratel (PT Dayamitra Telekomunikasi) is a subsidiary of Telkom Group and a sister company to Telkomsel, Indonesia's largest operator. Its ambitions of being the largest digital infrastructure are becoming ever realised – in December 2023 the company acquired some 803 towers from independent towerco PT Gametraco Tunggal for US\$112 million. 70% of those towers are situated outside the main island of Java. Its expanding towerco portfolio is now complemented by extensive fibre cabling. Mitratel acquired 9671km of fibre optic cable also in December to help fulfil its commercial model with a tower business that can achieve a billable ratio of 1.18x.

The company has also grown inorganically, and as well as acquiring sites from its sister company. In August 2022, Mitratel signed a deal to acquire 6,000 towers from the operator, and also completed a 2019 tower deal with Indosat Ooredoo. Additionally, in 2019, they also acquired fellow towerco PT Persada Sokka Tama (PST) and its 1,589 sites.

Mitratel completed their IPO in November 2021, raising US\$1.32bn. This included a US\$800m commitment from the Indonesia Investment Authority.

In February 2021, Mitratel announced plans to invest US\$69.5mn of the capital raised from the IPO in rural tower sites in the country. In February 2023, Mitratel acquired 997 towers from IOH, which came from the Hutchinson side of the business. Tower Bersama is listed on the Indonesia stock exchange (IBX: TBIG) with two of its portfolio companies Gihon (IBX:GHON - in which Tower Bersama has a 50.43% stake) and PT Visi Telekomunikasi Infrastruktur (IBX:GOLD - in which Tower Bersama has a 51.09% stake) separately listed.

Tower Bersama Infrastructure was formed back in 2004 by the rollup of several Indonesian towercos, and have completed several other acquisitions since. Tower Bersama is listed on the Indonesia stock exchange (IBX: TBIG) with two of its portfolio companies Gihon (IBX:GHON - in which Tower Bersama has a 50.43% stake) and PT Visi Telekomunikasi Infrastruktur (IBX:GOLD - in which Tower Bersama has a 51.09% stake) separately listed. Tower

Bersama's most recent tower deal involved the acquisition of 3,000 towers from IBS Tower, which was completed in April 2021. Following that deal, Tower Bersama reported just over 19,000 owned towers. In May 2022 Australian asset manager, Macquarie, and Canadian pension fund, PSP Investments, spent a combined US\$610 million to acquire a minority stake in Bersama Digital Infrastructure, who in turn own a 64.5% stake of Tower Bersama.

This fresh investment was followed by Bersama Digital's first expansion beyond Indonesia. The company partnered with Filipino investment firm Opti-Teknologi to acquire a majority stake in Alliance Tower Corporation (A Philippines-based towerco) for an undisclosed fee.

EdgePoint Infrastructure was founded in partnership with former EDOTCO CEO, Suresh Sidhu and US investment firm DigitalBridge. In early 2021, EdgePoint Infrastructure announced that it had reached a deal with Indosat Ooredoo to acquire 4,247 towers for US\$750mn, the deal having closed in May. This sale marked the final release of towers from Indosat Ooredoo. EdgePoint Infrastructure acquired Centratama in a series of equity transactions throughout 2021 and In March 2022, Centratama purchased the shares of EdgePoint's Indonesian operation to consolidate their operations in the country.

American Tower are also exploring entry into Indonesia, yet no formal plans are currently in place. Despite the wealth of towerco activity, there is still a consensus that Indonesia's networks are underserved.

Japan

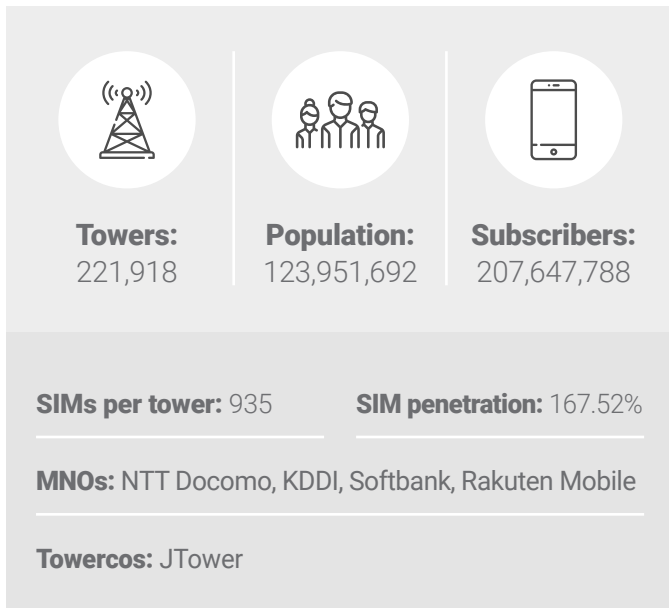
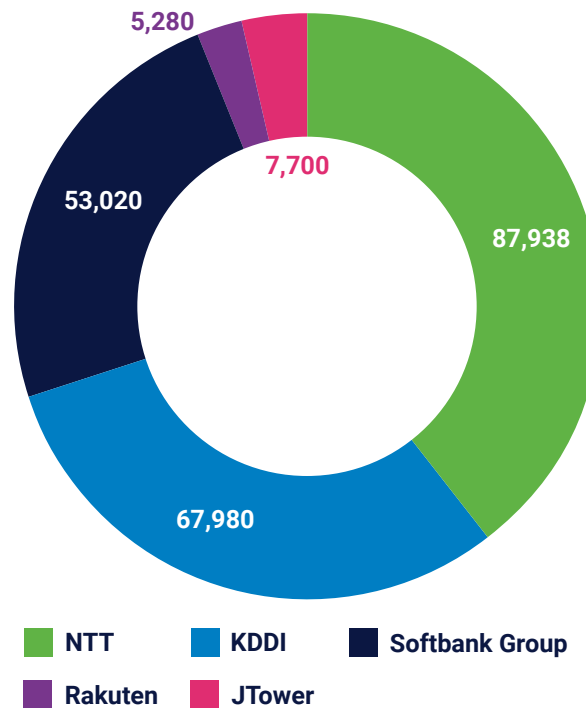


Figure 12: Japan - estimated tower ownership



Source: TowerXchange

Management signed a sale and leaseback transaction covering a portion of Rakuten's mobile assets in Japan. It is believed that Rakuten will continue to manage its estimated footprint of 5,280 towers, with the ownership of the towers reverting at the end of the

financing term. Digital infrastructure asset investors have recently been taking a keen eye on Japan, and it is apparent why.

Japan is one of the world's most advanced mobile markets in term of 5G rollout, with four MNOs operating – namely DOCOMO, KDDI, SoftBank and new market entrant Rakuten (Rakuten's network being based upon Open RAN technology). Rakuten reported record losses as a result of heavy network rollout costs but has vowed to increase network investment by 30-40% as it looks to grow its subscriber base and compete with the country's established players.

Calls by Japan's Prime Minister to reduce mobile fees is expected to intensify competition further. In a bid to reduce prices, the government has published guidelines around infrastructure sharing, of which there has been little to date.

Despite having previously mainly focused on indoor sites due to the aforementioned reluctance to share on towers, JTower acquired 71 towers from NTT in July 2021 for an undisclosed sum. In March 2022, JTower entered into a master transaction agreement to transfer a further 6,002 towers for a sum of US\$869.6 million. The first 555 of these sites were transferred to JTower in early 2023.

Laos



Towers:

9,613



Population:

7,529,475



Subscribers:

4,822,973

SIMs per tower:

502

SIM penetration:

64.05%

MNOs:

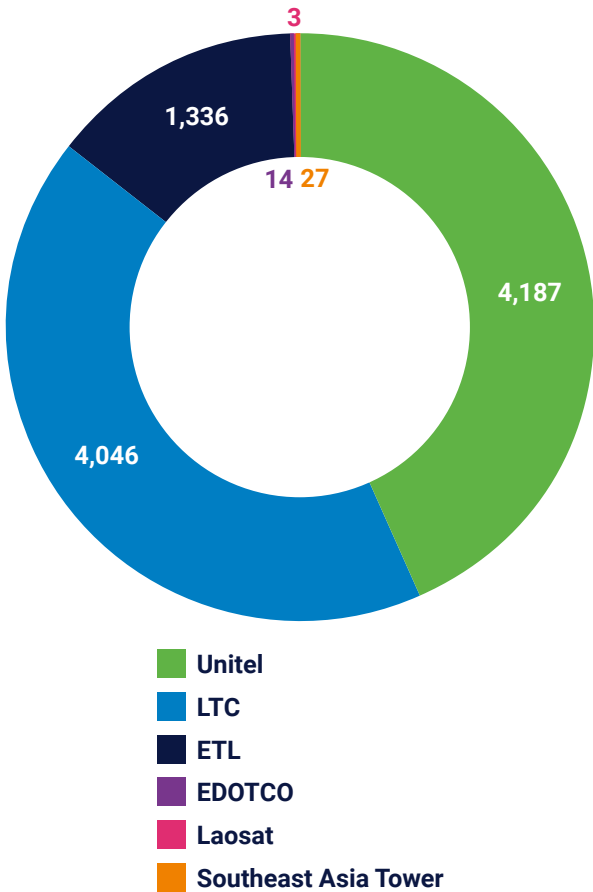
TPlus, ETL, Lao Telecommunications Company, Unitel

Towercos:

Southeast Asia Tower Company, EDOTCO, OCK Laos TowerCo

Laos is a relatively new market for towercos, with the national government and local firm Click Lao Marketing and Consultancy signing an agreement with China Tower Corporation (CTC) to establish the Southeast Asia Tower company in 2018. In 2019, EDOTCO presented a filing at the Bursa Malaysia announcing its entrance into Laos via the purchase of an 80% stake in local entity Mekong Tower Company Ltd. (MTCL).

Figure 13: Laos - Estimated tower ownership



Source: TowerXchange

However, EDOTCO are yet to build any towers since Q4 2020, meaning they still only own 14 towers nearly four years after entering the country.

Things appear to be on the up for the country though, as Malaysian tower builder OCK Group signed a shareholder agreement with Laos Ministry of Finance to expand its presence in the country. OCK Laos TowerCo will be a joint venture between the two parties, with OCK owning a controlling 70% stake.

In May 2024, OCK marked its entry in the Laos tower market, having signed a tower lease agreement with Best Telecom. OCK will build and lease telecommunication towers to Best Telecom in major cities in Laos over a 15-year period. OCK will support Best’s 5G rollout in the country with the launch of the network targeted for by the end of 2024, after having been awarded 5G licence in the country.

The country is home to five MNOs, three of which are owned by the government. TPlus, ETL and LaoTel, are joined by Viettel back Unitel and Best. The market holds great potential for growth with only 80% SIM penetration with 5.6mn mobile connections and 7.2mn inhabitants. Some degree of sharing exists within the government owned operators as TPlus have access to 1,000 of LTC’s towers.



Malaysia



Towers:
34,650



Population:
33,938,222



Subscribers:
47,952,100

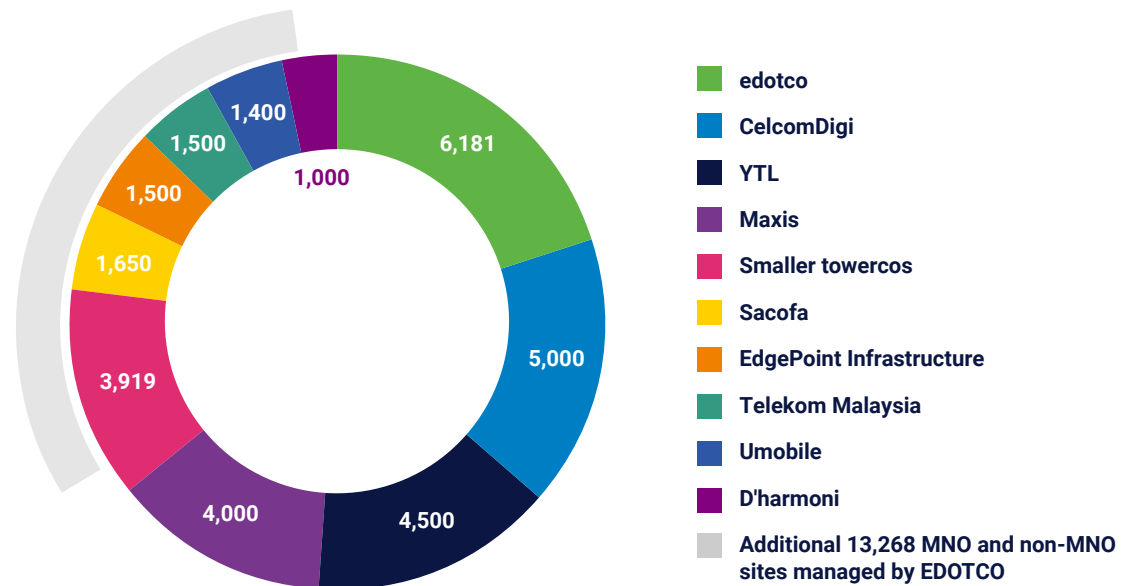
SIMs per tower: 1,382

SIM penetration: 141.29%

MNOs: Celcom Digi (Telenor & Axiata), Maxis, Telekom Malaysia, Umobile, Yes (YTL), ALTEL

Towercos: EDOTCO, Sacofa, EdgePoint Infrastructure, D'harmoni, OCK, KJS, Infraquest, Naza Communications, Desabina, Common Tower, Tele-flow, Perak Integrated Networks, Omnix Malaysia, Rangkaian Minang, PDC Telecommunications, Bullish Aim, Redpyne, Stealth Solutions

Figure 14: Malaysia - estimated tower ownership



Source: TowerXchange

Malaysia's unique approach to 5G has taken another step forward. In October 2024 the Malaysian government announced the award of its second open access 5G network to the country's smallest MNO, U Mobile. Digital Nasional Berhad (DNB) will continue to operate the first open access network as it has since May 2023. Four MNOs - Maxis, U Mobile,

YTL and CelcomDigi – have taken an equal 16.28% share in DNB, leaving the Ministry of Finance with an approximate 35% stake in DNB). With two networks rolling out 5G TowerXchange expects to see lease up increase across the networks of the towercos active in Malaysia, following the depressed levels of co-location over the last years.

The country is home to a competitive MNO market, led by multinational MNOs Maxis and Celcom Digi; the latter was formed following a merger between Telenor and Axiata's businesses in the country. As summarised at TowerXchange's Meetup Asia 2023 by Malaysia's Minister of Communications, Fahmi Fadzil, policy in Malaysia is driven by digital transformation, facilitated



MALAYSIA

by 5G densification and plans for complete 4G coverage across the country. The tower industry is seen as a key enabler of cutting-edge high speed connectivity in densely populated urban hubs as well as closing the huge remaining rural infrastructure gap, said Fadzil.

The country's largest tower, EDOTCO, has pursued a policy of rolling up state backed towercos in recent years, and DigitalBridge backed towerco EdgePoint Infrastructure has begun following suit after commencing operations in late 2020.

Service provider OCK own approximately 500 towers in the country, and other non-state backed towercos include Tele-flow, Naza Communications and Omnix.

EDOTCO's has been active on the M&A front with the acquisition of local towerco Touch Mindscape. The acquisition boosted EDOTCO's tower count by around 1,000 sites (89% of which are ground based towers) and they also acquired approximately 750km of fibre.

DigitalBridge backed EdgePoint Infrastructure, headed by former EDOTCO CEO, Suresh Sidhu, entered the Malaysian market in February 2021 with the acquisition of small local player GTP Network. EdgePoint bolstered their position in June with the acquisition of slightly larger towerco, Asiaspace.

In 2021, Umobile was rumoured to be considering both an IPO, and a sale of its 1,400 sites for a sum of

around 500 million ringgit (US\$120 million). However, neither transaction occurred. It is now rumoured that U Mobile will move forward with an IPO of at least RM10b (US\$223mn) in 2025. Winning the second 5G network contract will increase U Mobile's capex and therefore may also lead to a renewed interest in selling its towers

Operators in Malaysia still own over 16,000 sites in Malaysia. Many of these sites are managed by the operators themselves, but Malaysia is also one of EDOTCO's largest managed sites markets, and the towerco collects a fee to manage 13,268 telecom towers in the market.

While The Malaysian Communications and Multimedia Commission (MCMC) has offered well-defined guidelines at a national level, regulatory complexity exists at a state level. State backed towercos are solely responsible for permitting and building towers in certain states, which makes it challenging for national players such as EDOTCO.

Whilst it is a mature industry, the tower market is still growing with 1,000-2,000 new structures, including street level furniture, being put up every year. The government owned Jendela scheme has been responsible for funding around 2,000 new 4G towers since September 2020, mainly in rural areas that lack commercial viability.

The scheme was refreshed in the March 2023 budget

by the new government and will be expanded to provide digital connectivity in 47 industrial areas and nearly 3,700 schools with a budget of RM 750 million.

Growth is being driven by operators needing to support current 4G requirements, as well as meet the constantly growing demand for higher capacity and speed as data usage increases. Most of the growth opportunities for towercos come in the form of infill solutions.

Operational complexity in Malaysia is low, with towercos primarily focusing on the land and structure, while responsibility for power remains with MNOs. Load issues mean that battery deployment is widespread, but grid reliability is generally solid, especially in urban areas. Less populated states of Sarawak and Sabah are where the vast majority of Diesel Gensets in the country can be found.



Mongolia



Towers:
1,000



Population:
3.3mn



Subscribers:
4.4mn

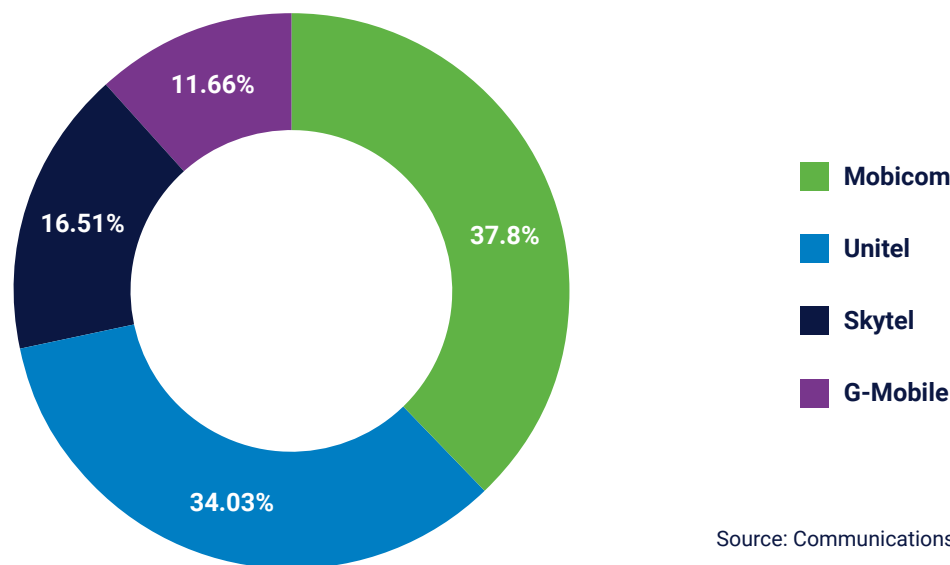
SIMs per tower: 4,419

SIM penetration: 134.8%

MNOs: G-Mobile, Mobicom (KDDI), Unitel, Skytel T

Towercos: none

Figure 15: Mongolia - MNO market share



Source: Communications Regulatory Commission of Mongolia

Mongolia is home to four MNOs – G-Mobile, MobiCom (KDDI), SKYtel and Unitel – and 4.4mn mobile connections with a population of 3.2mn. Almost half of Mongolia's population is located in the capital Ulaanbaatar, which homes 1.45 million people and the majority of the telecoms infrastructure. Following the separation of the telecom and infrastructure businesses in 2013, a few infrastructure providers now run towers,

active equipment, fibre and microwave backhaul across the 1.5mn sq km Mongolian landscape.

There are approximately 1,000 towers in the country, and more than half are shared between the operators.

In March 2021, MobiCom began 5G trials, but no spectrum has been allocated in the country.

Myanmar

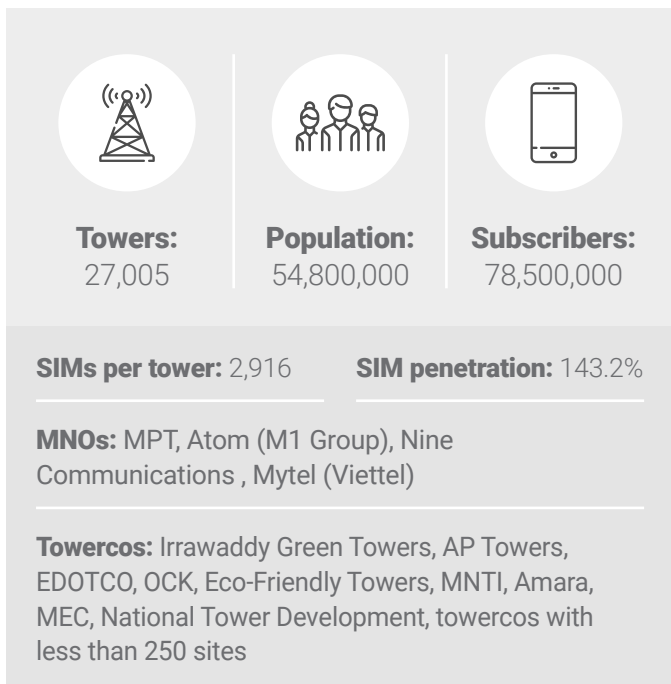
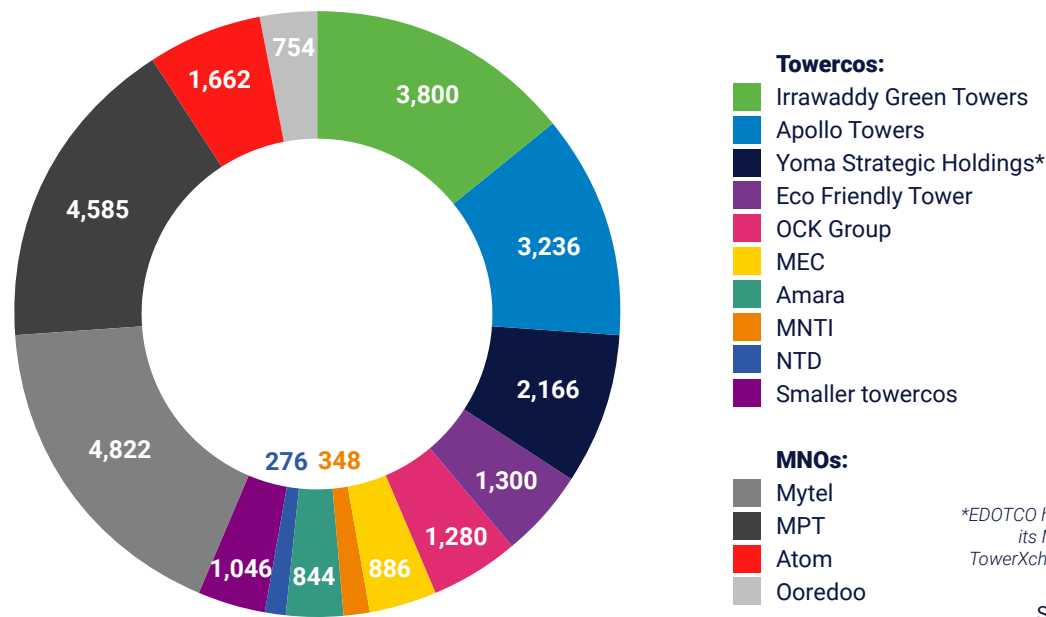


Figure 16: Myanmar - estimated tower ownership



Source: TowerXchange

The history of the Myanmar tower industry is a turbulent one. Just eight short years ago, there were no towercos on the ground and buying a SIM card was a prohibitive privilege. However, Myanmar's digital landscape began to change in the following years up until 2015 when entrants like Telenor and Ooredoo were granted licences and an independent regulatory body created to facilitate a competitive and healthy tower market.

In In 2013, Myanmar had one operator, less than 7mn phone users and 7,600 km of fibre. The new operators were supported by a wave of towercos that have built over 10,000 towers between them in just five years. But by the end of 2020, the country had in excess of 56mn mobile subscribers and a 110% mobile penetration.

The Post & Telecommunications Department (PTD)

reported 23,381 towers across the country's diverse geography with a tenancy ratio of 1.35, which varied from 1.1 to 2.2 between different players.

Even in January 2021 Swiss private equity giant CVC group agreed to purchase Irrawaddy Green Towers (IGT), the leading towerco in the country for a sum of US\$700m.



MYANMAR

Fast forward 6 months to July 2021 and the country risk became so untenable that the second largest MNO, Norway-based Telenor, announced their exit from the market with a sale to M1 group for a measly price of US\$105mn.

So what went wrong? Put simply, the military coup that occurred in February 2021 has crippled a resurgent Myanmar that was being held up as one of the golden children of ASEAN telecoms. The subsequent protests, and a brutal wave of Covid-19 has brought the country to its knees, with the new government going as far as shutting down the internet in April amid multiple accusations of human rights abuses

In an act of defiance against partly state-owned MyTel, local militia groups targeted telecom infrastructure throughout the latter half of 2021 and destroyed Mytel towers and co-locations, provoking the Junta to respond by planting landmines across their sites.

Towercos whose were mined immediately halted site maintenance visits, and crucially refueling, which led to inactivity and a lack of internet access for over one million people.

4G densification and 5G rollout plans in the form of investment in street furniture and smart poles have been all but scrapped.

Until everything was put on halt, Myanmar saw an

explosion of towerco activity. Staying true to their approach in other markets, Viettel entered in 2018 and have been developing many small infrastructure subsidiaries across the country to satisfy its growing BTS needs. Many of these entities are becoming acquisition targets for the bigger towercos.

It also relied on independent towercos as the company recognised it would not be able to achieve its coverage commitments without their support and expertise. Mytel have built over 3,500 new towers and co-located on over 5,000 sites.

After five years in the country, Mytel became the leading operator in terms of market share and service quality in this country with 13 million subscribers. It is the operator with the largest and best 4G coverage in Myanmar with an area of nearly 90% of the population.

All four operators had been testing 5G and regulator PTD estimated that the country could be launching the technology in two to three years.

This now looks increasingly unlikely. In the meantime, MNOs were planning on continuing to expand their networks through BTS and colocations on macro sites while also testing new solutions and typologies in urban environments. This has also been stripped back.

Grid power is still unreliable even in major cities and power is a major issue for the industry. More than 50%

of the sites are in off-grid locations hence towercos and MNOs rely on gensets and batteries while exploring the benefits of hybrid systems, renewables and lithium-ion batteries to power and cover their operations.

Despite a significantly challenging 2021, the market returned to some sense of normality in 2022. It is still considered “safe” to own and operate towers in Myanmar. Tenants are still paying their bills, and towercos are even being awarded permits to build new sites.

However, sanctions placed on and by the new government make moving money in and out of the country much harder. This means it is unlikely we will see many towers in Myanmar change hands outside of the existing companies.

In September 2022 Ooredoo sold its Myanmar operation, to Singapore-based Nine Communications in a deal that valued the business at \$US576 million.

Pan-Asia towerco EDOTCO is selling its 87.5% stake in its Myanmar towerco to a local investment group, Yoma Strategic Holdings, and relinquished 3,073 towers it had in the country as part of the sale, according to the latest set of financial results of its owners Axiata. Axiata announced that a US\$150mn sale had been agreed in a filing with the Malaysian stock exchange in April 2024.



Nepal

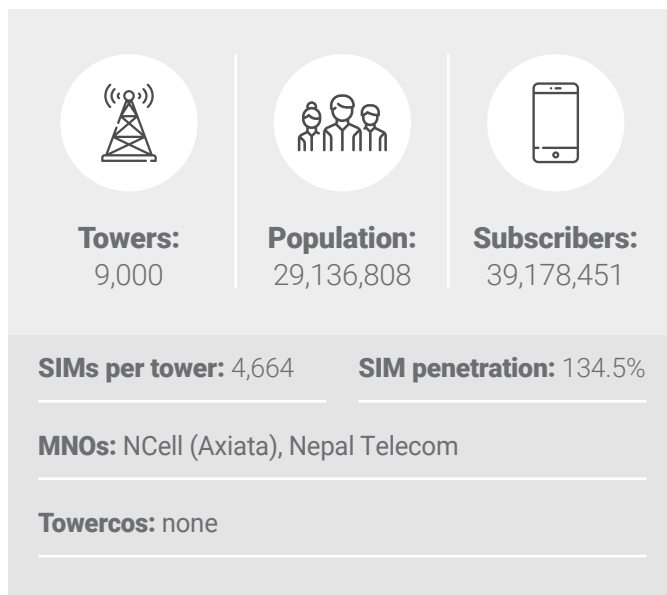
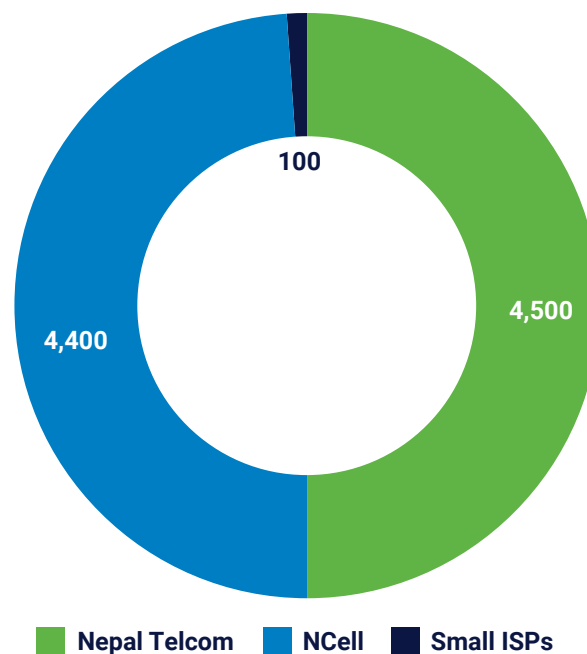


Figure 17: Nepal – MNO market share



Source: Nepal Telecommunications Authority

Two MNOs had operated in the Nepalese market, state-owned Nepal Telecom, and Axiata-owned NCell. In 2019, the Nepal Telecommunications Authority (NTA) threatened to issue an MNO licence to an international player due to concerns about the lack of competition and investments in the telecoms market. However, due to a lack of suitable candidates, the idea was scrapped.

No tower sharing exists between the two mobile operators, despite regulatory pressure. The regulator

would like to issue a licence to a third mobile operator and encourage the entrance of independent towercos to inject new capital in to the market, but the market has limit appeal to international operators.

A few years ago, the NTA published a draft Infrastructure Development and Sharing Regulation, seeking request for proposal from towercos to provide telecom infrastructure services. Although at least eight international firms were interested, the Ministry of Communications and Information Technology (MoCIT) never issued any licenses and haven't to this day.

In July 2021 a minor by-law was passed that prohibited a service provider from rejecting or delaying a request for site access from another company. The right to use this infrastructure will be terminated if they are not used within 3 months of the agreement being signed. The law extends to cover fibre ports, dark fibre cabling, wireless network point to point data and base transceiver stations and others.

Nepal has one of the lowest levels of access to electricity in Asia, at under 90%. 88.8% of rural locations are off grid, and power availability in urban areas is still only at 94.2%. Only Myanmar has worse access to electricity for cell sites, meaning energy related issues will be particularly important once towercos break into Nepal.

There's a huge requirement for new infrastructure, in particular in rural areas. 2G has only reached 90% coverage, but because of the low margins both operators are hesitant to committing to rural rollout plans.



New Zealand



Towers:
4,799



Population:
5,185,288



Subscribers:
5,947,000

SIMs per tower: 1,239

SIM penetration: 114.69%

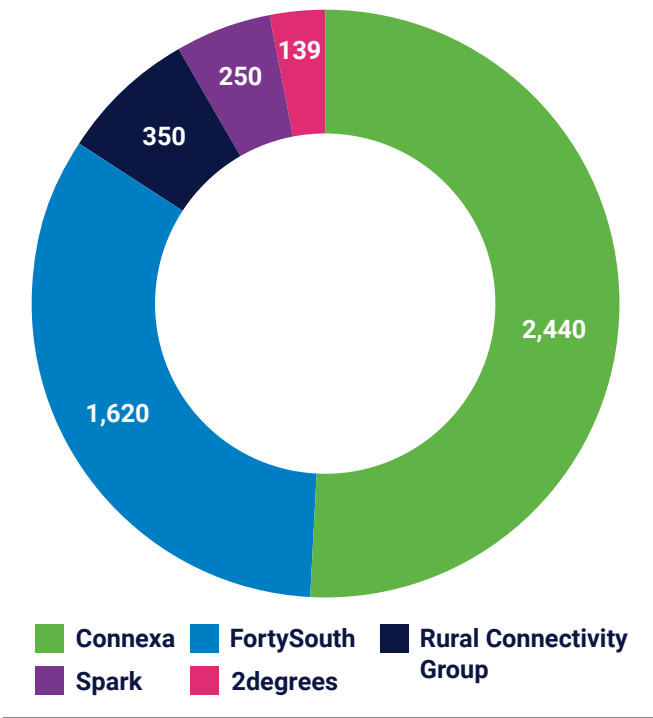
MNOs: Spark NZ, Vodafone NZ, 2degrees

Towercos: Connexa, FortySouth, Rural Connectivity Group

In July 2022 the New Zealand tower industry underwent a dramatic transformation in the space of just a week. Following announcements from incumbent MNOs Spark and Vodafone NZ back in February that individual processes to find external capital in tower assets were being explored, both MNOs announced their successful buyers hot on the heels of one another.

The first of the two deals saw “TowerCo”, the tower carve-out of Spark New Zealand, sold to Ontario

Figure 18: New Zealand - estimated tower ownership



Teachers’ Pension Plan (OTPP) for US\$900mn. OTPP will acquire a controlling 70% interest in the TowerCo business, now called Connexa.

Spark carved out 1,263 sites into its towerco and has retained around 250, and instead concentrating on

reinvestment into expanding its data centre portfolio. In line with other transactions across the Tasman Sea in Australia, the transaction valued Spark Towerco at \$1.1 billion, for an FY23 Pro Forma EBITDA multiple of 33.8x.

Spark has entered into a fifteen-year agreement with Connexa as anchor tenant and the new independent towerco will have a build commitment of 670 new sites to be completed over the next ten years. Although only tower assets have been sold, under OTTP, Connexa will support New Zealand’s MNOs with a full suite of digital infrastructure services and assets. Now New Zealand’s largest towerco, Connexa acquired ATC New Zealand, trading as Clearspan, in 2024. Clearspan is a land aggregator.

The formation of Foreysouth was led by global asset management firms InfraRed Capital Partners and Northleaf Capital Partners each acquiring 40% of Vodafone New Zealand’s 1,484 towers. Vodafone New Zealand is owned by Brookfield and Infratil rather than its UK based namesake, with Infratil planning to reinvest proceeds from the sale to acquire the final 20% of the business.

A small number of sites are owned by a consortium rolling out rural sites in the large but lightly populated country.



NEW ZEALAND

Vodafone New Zealand will enter into a 20 year agreement with the new towerco to maintain access to the towers, and the deal also includes an additional 390 sites to be rolled out over the same 10 year period as Spark.

In a strikingly similar deal, the transaction and has raised US\$1.1 billion for Vodafone New Zealand, for an FY23 pro-forma EBITDA multiple of an identical 33.8x. Completion of the transaction is subject to Overseas Investment Office approval.

Exceeding Spark TowerCo's site count, the new towerco will be the largest independent owner of tower assets in New Zealand, and will cover 98% of the country.

InfraRed Capital Partners is an international infrastructure investment manager with over US\$12bn of equity capital in multiple private and listed funds. The acquisition marks InfraRed's third investment in New Zealand, and its first globally in telecom towers.

Northleaf Capital Partners is a global private markets investment firm with over US\$20 billion in private equity, private credit and infrastructure commitments. Northleaf has been an active investor in Australia and New Zealand infrastructure projects since 2013, but again this marks the first move into telecom towers for the Canadian based fund.

In December 2022 Connexa completed the set of

New Zealand's tower portfolio sales by buying 1,124 mobile towers from 2Degrees for US\$1.08 bn. Under the terms of the deal, 2degrees has entered into a 20 year-agreement with Connexa (plus rights of renewal) to secure access to existing and new towers, with an additional tower build and co-location commitment of 450 sites over the next 10 years.

Fortysouth is New Zealand's other established towerco, with a site count of 1,590 – around 950 of which are freestanding. It operates in a market where infrastructure is advancing in order to facilitate the country's exponential growth in data. Towers are being strengthened in order to accommodate 5G and to harness colocation. And with 85% of homes in the country having fibre connection, there is significant data growth being driven by fixed wireless.

There were early signs of a nascent tower industry emerging in New Zealand, where Spark and Vodafone New Zealand have substantial but ageing tower networks. Newer entrant 2degrees have leveraged co-location where possible and alongside building their own sites have extended their coverage to around 95% of the country.

2degrees may also have an appetite to sell their towers and partner with a towerco on BTS. Parallel infrastructure is substantial, while the need for improved rural coverage, particularly on the South Island where tourist and agribusiness drive demand, has prompted

the government's Rural Broadband Initiative (run by Crown Infrastructure Partners) to invest in over 100 towers - particularly along transport routes.

CIP have since extended network coverage to over 2,000km of highway and over 170 tourist destinations.

Multinational towerco American Tower's New Zealand property and land business was exchanged hands to Connexa in an M&A deal in September 2024. Some 112 sites including land under mobile towers and rooftop based mobile sites are now at the hands of the towerco.

A total of just under 4,500 macro-towers are supplemented by around 7,000 rooftop sites, primarily used in the larger cities. In early 2021, Vodafone New Zealand announced a "turbocharged" digital infrastructure investment programme.

The milestone 250th tower was completed as part of this project in June 2021, and was built by the Rural Connectivity Group. The Rural Connectivity Group is a joint venture run by all three of the nation's MNOs to develop rural connectivity and eliminate mobile blackspots.

The Rural Broadband Initiative has also committed NZD10 million (USD7.06 million) to deploy suitable radio spectrum in more rural areas. It is expected that this will vastly improve broadband capacity and coverage in the long-term, as the 600MHz band is better suited to 5G.



Philippines



Towers:
37,197



Population:
115,559,009



Subscribers:
166,454,458

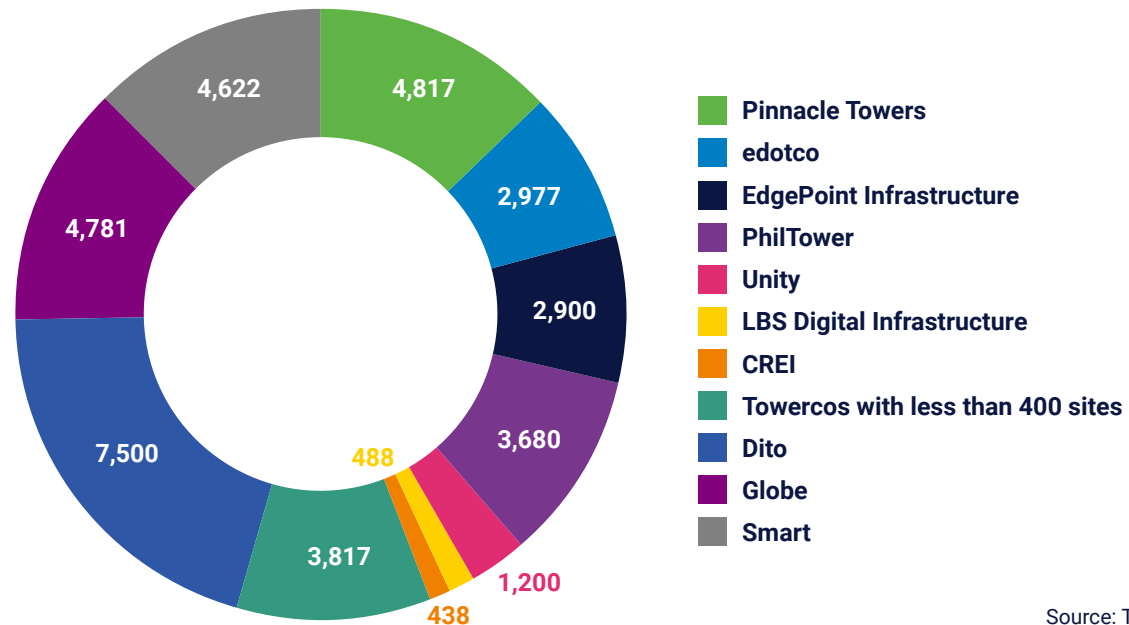
SIMs per tower: 4,473

SIM penetration: 144.04%

MNOs: Smart (PLDT), Globe (Singtel/Ayala), Dito

Towercos: Towercos that were awarded licences included: IAcoda Towers SDN BHD, Alt-Global-Solutions Inc., Transcend Towers Infrastructure (American Tower), China Construction First Group Corp. and Wigan Construction and Development Corp., China Construction Yangtze River, China Energy Equipment, CREI Management Services FZE, Desarrollos Terrestres, EEI Corp, Frontier Tower Associates Philippines, IHS Holding, Inforient, ISOC EDOTCO, ISON Tower, J.S Cruz Construction and Development, LBS Towers, MGS Construction, Phil-Tower Consortium, RT Telecom, SBA Communications, Shinheung Telecom Co, Tamoin Industrial Services Corp., Tiger Infrastructure, UA Withya Public Company Ltd

Figure 19: Philippines - estimated tower ownership



Source: TowerXchange

The Philippine tower market has undergone a significant transformation in the past four years, most recently with the formation of a brand-new joint venture that intends to transform the 4G and 5G mobile and digital infrastructure in the market. In February 2024, independent towercos PhilTower and MIDC announced a brand-new JVco that will combine and consolidate its presence in key Filipino markets in Luxon, Visayas and Mindanao, whilst

bringing improved coverage for MNOs and supporting the country's connectivity needs. Macquarie Capital is now exploding a sale in the joint venture which may value the new company at US\$700mn to US\$800mn. Prior to the merger MIDC was busy on the transaction front, most recently completing the purchase of 154 additional towers from Globe in March 2024, with 62% of Globe's 2,180 towers in a deal worth US\$468mn now in the hands of MIDC.



PHILIPPINES

Tasked with increasing coverage across the challenging Filipino landscape that includes over 9,000 islands and 100 million people, Dito has got off to a flying start, exceeding targets by amassing 12mn subscribers and over 7,500 towers by the start of 2023.

In addition to the establishment of a third operator, a common tower policy was established that supported objectives to significantly increase coverage and capacity. Industry experts estimate that the Philippines requires 70,000 sites to hit its coverage goals.

A new favourable regulatory framework, immense build to suit opportunities and the prospect of significant SLB deals with Smart and Globe has spurred huge interest in the country. However, independent tower companies (or common tower companies as they are known locally) in the Philippines must be at least 20% owned, or in a consortium with, companies with at least five years of experience constructing, owning, operating and/or maintaining towers in the country.

The first of these deals was announced in April 2022 with Malaysian-based towercos EDOTCO and EdgePoint each acquiring 5,907 towers between them for just under US\$1.5bn. Globe's first transactions were announced in August 2022 with KKR backed Frontier Tower Associates, and MIDC, a telecom tower infrastructure subsidiary of Meralco.

Frontier Tower successfully acquired 3,539 towers for

US\$810mn and signed an agreement to secure Globe access to the towers for an initial period of fifteen years.

Smart announced two more transactions in December 2022 and March 2023 with Unity and Frontier respectively. Unity secured their second sale and leaseback, this time with Globe, in May 2023.

Local conglomerate Aboitz Infrastructure created Unity in partnership with international private equity player Partners Group.

In June 2022, ISON Tower secured funding from Peppertree Capital, who have also invested in Australia and US towers recently.

I-eng's investment arm, CREI, have also launched a towerco in the Philippines, building 250 sites and securing investment from the IFC. Bersama Digital have also made their first move outside of Indonesia by investing in Alliance Towers Corporation and Singapore headquartered Tiger Towers have also built around 100 sites.

Despite a boom period brought around by the initial licencing of new towercos and the first sale and leasebacks, financial irregularities have curtailed digital infrastructure development in the Philippines. Firstly, DITO are experiencing financing issues.

DITO were required to cover 70% of the Philippines by July 2022, a target which they hit, but preferred to keep

their tower deployment in-house to reduce the risk of their licence being revoked if the new towercos were not up to the task. DITO has agreed a handful of co-locations and a small number of BTS sites with towercos, but is responsible for just a small fraction of towerco deployments.

Secondly, Smart's parent company PLDT has been rocked by a probe into a US\$886mn budget overrun. The operator could not account for 12.7% of its capex investment in the past four years, which resulted in the senior leadership team being suspended or leaving the company.

To round things off, Globe made the decision to focus more on co-locations than new sites in 2023. Globe built around 2,000 towers a year between 2020 and 2022 with towerco partners, but have reduced this to just 500 in 2023.

Key challenges the Philippines faces are around permitting and securing grid connections to power sites. Permitting laws in the Philippines are incredibly outdated, and face numerous local nuances.

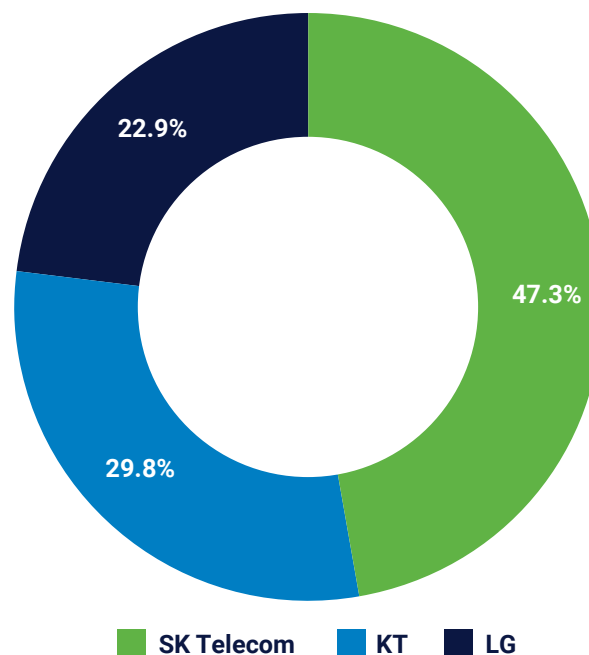
From an energy standpoint, The Philippines grid is managed by several local power co-operatives, who are responsible for generating and supplying electricity in their dedicated region. Some of these co-operatives are small organisations that have manual workflows and severe red tape that is making electrification of tower sites challenging for towercos. This is especially the case for international players who are new to the country's local customs.



South Korea

		
Towers: 30,000	Population: 51.8mn	Subscribers: 68.9mn
SIMs per tower: 2,296		SIM penetration: 133%
MNOs: SK Telecom, KT, LG		
Towercos: None		

Figure 20: South Korea – MNO market share



Source: The Korea Information Society Development Institute

According to data published by the Ministry of Science and ICT (MSIT) there are a total of 30 million 5G subscription in South Korea, translating to just over 43.5% of all mobile accesses in the country. Each of the three MNOs are said to have installed between 60,000 and 70,000 5G base stations over the past two and a half years, resulting in a total count of 200,000 across the country.

South Korea's three mobile network operators will allow shared access to their respective 5G networks in remote coastal and farm towns. TowerXchange is starting to pick up the first faint signals that towerco activity may be emerging in South Korea.

South Korea boasts one of the most sophisticated telecommunications infrastructures in the world, cultivating an insatiable demand for high-speed mobile broadband among its citizens. Mobile broadband penetration in South Korea is above 99% and fibre has been widely deployed. South Korea is a three-operator market featuring SK Telecom, KT and LG Uplus.

The Ministry of Science, ICT and Future Planning (MSIP)

has tried multiple times over the years to license a fourth MNO, however, failed again in February 2017 as none of the three applicants (Sejong Telecom, K Mobile, and Quantum Mobile) met the criteria.



Sri Lanka

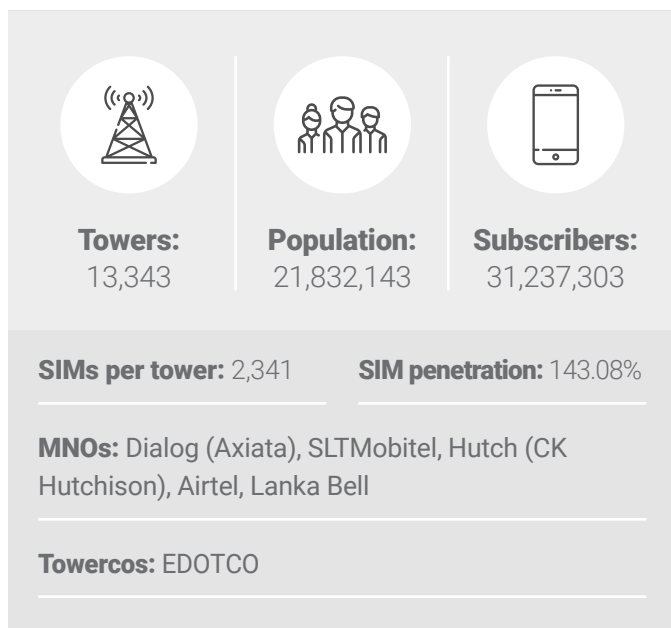
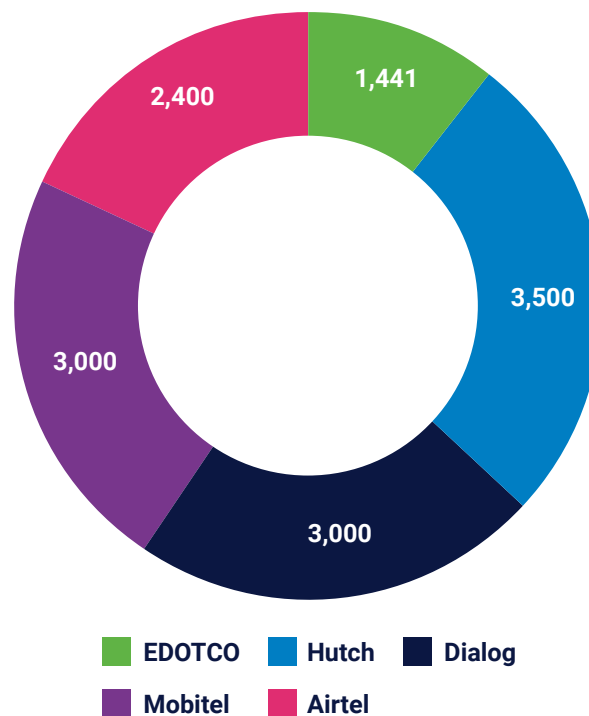


Figure 21: Sri Lanka - estimated tower ownership



Source: TowerXchange

There are five MNOs in the Sri Lankan market, Dialog (owned by Axiata), state controlled SLTMobitel, Lanka Bell, Hutch (owned by Hutchison and which acquired Etisalat Sri Lanka in 2018) and Airtel (part of Bharti Airtel Group).

Consolidation could be on the cards in the market amidst intense competition, high taxes and high capex requirements for the sector. Dialog and SLT-Mobitel hold all of the 4G spectrum, and any other players that want to offer this will need to engage in RANsharing.

Hutch Lanka teamed up with ZTE and achieved 1.8Gbps data transmission in its trial in March 2021. The TRC has implemented an initiative dubbed 'Gamata

Sanniwedanaya', which seeks to eliminate the digital divide in the country, providing equal access to mobile connectivity for all Sri Lankans.

In April 2021, Airtel announced that it is deploying 37 new sites as part of the initiative. Dialog have also successfully built 515 towers since the start of 2019, which increased the total number of their towers to around 4,000.

There are approximately 10,000 towers in the Sri Lankan market with significant bilateral sharing between operators. EDOTCO is the sole towerco operating in the market.

A proposed legislation change from Sri Lanka's regulator could set to see an independent tower model flourish in the long term. Sri Lanka's Telecommunications Regulatory Commission (TRC) Act changed for the first time, with amendments passed in August 2024. The move is a significant for the industry as it will now allow independent tower companies, small cell and fibre operators, as well as neutral party host networks to start building telecom sites in the country.

Sri Lanka is home to approximately 7,000 mobile towers at the moment, but they can only be built by the nations telcos, in partnership with government parties.



Thailand

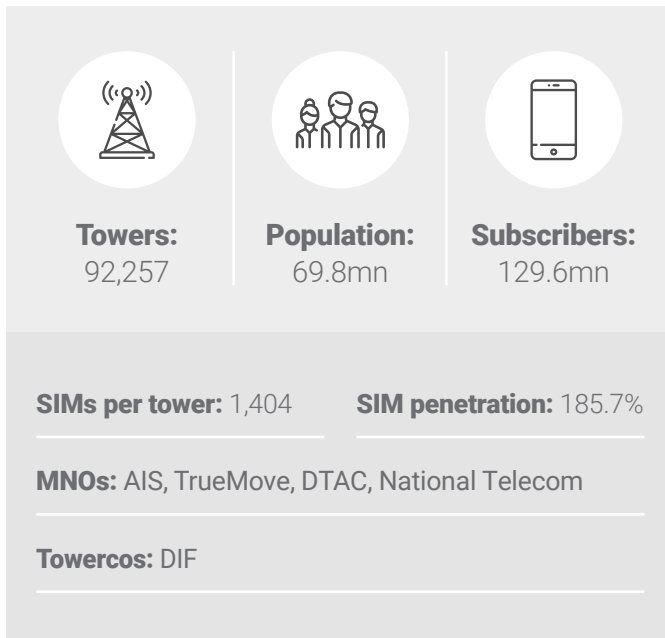
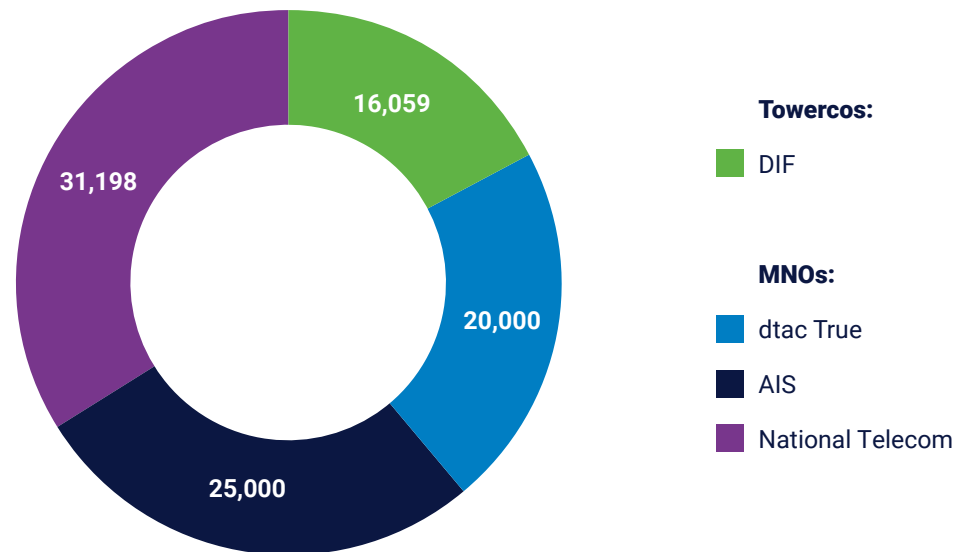


Figure 22: Thailand - estimated tower ownership



Source: TowerXchange

Ownership of Thailand's telecom towers has been an incredibly complex subject as a result of Build Transfer Operator (BTO) agreements between the nation's three private MNOs and state-owned entities TOT and CAT, which have now merged to form National Telecom.

The only non-MNO entity to own Thailand's towers is the Digital Infrastructure Fund (DIF). Between the IPO

and 2015, TRUE and its subsidiaries AWS and BFKT transferred ownership and the rights to revenue from 11,845 towers into DIF, alongside over 1 million km of fibre optic cable and transmission.

TRUE transferred further towers to DIF in 2015, 2017, 2018 and 2019 but are yet to transfer any of the towers they have built since, due to the pandemic. It is estimated that TRUE have built around 4,000 towers in this time.

Parallel infrastructure in Thailand is significant, and it is not uncommon to see three towers right next to one another, each with a single tenant. DIF has signed an agreement with DTAC to rent space on the towers they own on behalf of TRUE, but DTAC are currently only present on 2-3% of sites. The remainder of the portfolio is solely occupied by TRUE.

Rumours have been swirling in the market that AIS



THAILAND

has held talks with potential advisers for a strategic review of its assets, which could fetch the MNO between US\$2-3bn.

In early 2020, EDOTCO signed an agreement with CAT and a network infrastructure provider to collaborate in the development of shared network infrastructure for 5G rollout, although there has been little news on the Malaysian-based towerco's activities since.

Local firm ALT Telecom attempted to pioneer a tower sharing model in Thailand, signing a deal with a petrol station owner to build sites on their land. However the initiative was met with contempt from the country's MNOs, who preferred to build their own towers on adjacent land rather than rent existing towers.

A merger between DTAC and TRUE completed in early 2023, and could see consolidation in the tower

space. It is expected that around 5,000 towers will be decommissioned, due to crossover in the two MNOs existing networks, and no towers will be sold until one year following the completion of the merger.

While DIF and National Telecom own around half of the towers in the country, all operations and maintenance duties remain with the Thai MNOs. Fortunately, Thailand is a relatively straightforward market to operate towers.

Vietnam



Towers:
89,268



Population:
97.3mn



Subscribers:
136.2mn

SIMs per tower: 1,530

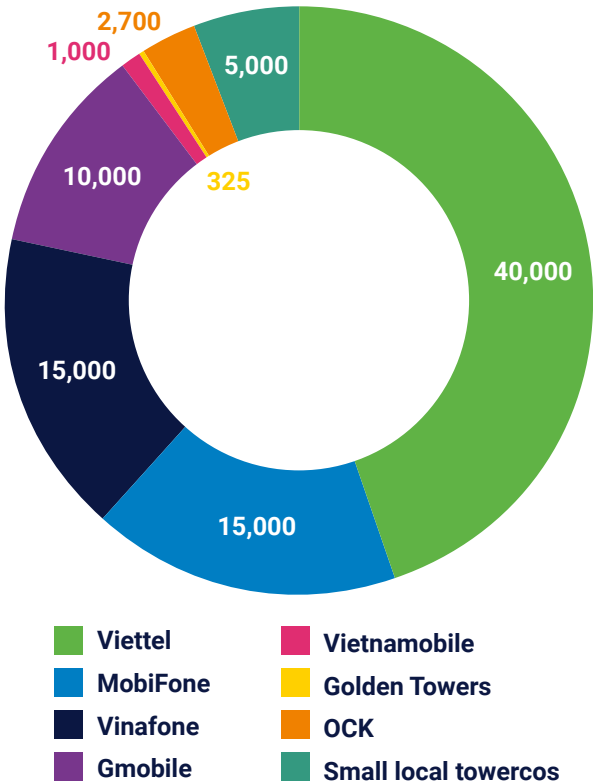
SIM penetration: 140%

MNOs: Viettel Mobile, MobiFone, Vinafone (VNPT), Vietnamobile, Gmobile

Towercos: OCK, Golden Towers, Small local towercos

Vietnam is home to four main Mobile Network Operators – Viettel mobile, MobiFone, Vinafone (VNPT) and Vietnamobile who between them share over 99.5% of the country’s estimated 136mn mobile subscriptions. The market is dominated by military backed Viettel, who account for 52.4% of network subscribers. VNPT and MobiFone follow with 24.18% and 19.59% respectively, while Vietnamobile has a 3.61% market share. The remaining 0.42% is shared between Gmobile and MVNO Indochina.

Figure 23: Vietnam - estimated tower ownership



Source: TowerXchange

Despite being a relatively mature telecoms market, Vietnam is proving that it still holds room for

significant growth. The nation of nearly 100mn people’s digital economy reached a total value of \$14 billion in 2020, which marked a \$2 billion increase from its valuation in 2019. In the same period, Vietnam was the country with the highest rate of new Internet users in the ASEAN region.

Vietnam’s operators have typically shied away from infrastructure sharing, and towercos have acted more as supporting actors than taking a lead role. For example, Viettel has control of its entire value chain, from tower manufacturing to energy The country’s tenancy ratio is estimated to be 1.5x, and there are no regulations that prevent MNOs building towers in the same location, which has caused significant overlap.

4G rollout has been relatively successful, and it is estimated that 30mn Vietnamese SIMs have access to 4G. However, MNO profits have not been as high as expected considering the latest wave of commercial opportunity, and the earliest forms of 5G seems at least a year off.

With the requirement to operate 5G networks profitably, Vietnam’s operators could be more seriously considering using towercos as a more significant commercial partner in the near future.

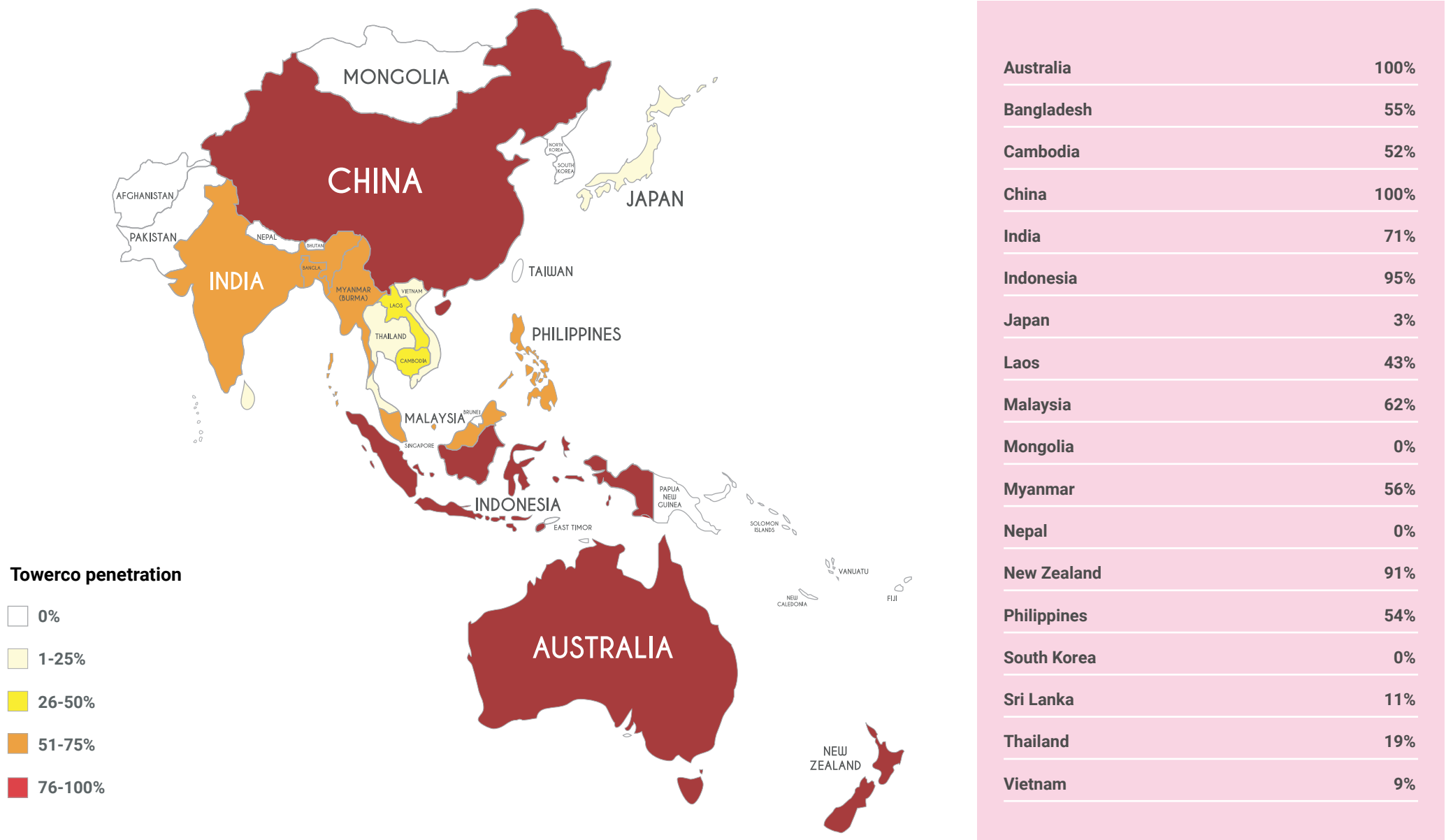
**Figure 24:** Heatmap of towerco penetration across select Asian markets



Figure 25: Expected M&A activity across the Asian tower market



